HERMITIAN–YANG–MILLS DYNAMICS AND HARMONIC MODULI NEAR A CONIFOLD

BERND JOHANNES WUEBBEN

Abstract. We construct a nonflat rank-four bundle family on compact Calabi–Yau threefolds smoothing eight ordinary double points and compare its Hermitian–Einstein metrics, metric flow and local moduli geometry uniformly through degeneration. A two-arrow matrix form determines all seven small trace-free endomorphism eigenvalues with relative error, including changing kernels and arbitrarily separated positive rates. On each polystable member, an open spatial C^1 basin admits a finite invariant graph and an initial phase for which the actual Hermitian–Yang–Mills flow tracks an exact graph trajectory with exponentially decaying C^1 error for all time. At the Hermitian–Einstein equilibrium, the Hessian of twice Donaldson's functional restricted to the graph is relatively equivalent to the same matrix form in the original metric coordinates. We also identify the actual local bundle classes and compare their harmonic metric on every stabilizer stratum. For paths confined to this locus, the unique metric tangent cone at the split bundle is $(\mathbb{C}^2, 4|\mathrm{d}u|^2)/((\mathbb{Z}/2)^2 \rtimes S_2)$. All comparisons allow independent node and bundle scales.

1. Introduction

Near a polystable bundle, a finite-dimensional deformation space describes the loss of rigidity of Hermitian–Einstein metrics. When the underlying Kähler manifold also degenerates, this description requires estimates uniform in the geometric parameters. An additive error in a finite operator is insufficient if some eigenvalues are much smaller than others. Likewise, a stationary reduction alone does not determine the trajectory selected by a spatially varying initial metric or the harmonic metric on nearby bundle classes. This paper proves these three comparisons on one nonflat bundle family over compact conifold smoothings.

1.1. The geometric family and the principal theorem.

We use a polarized family X_p of Calabi–Yau threefolds whose central fibre has eight ordinary double points. The base is a small ball in the six-dimensional space

$$\mathcal{R} = \left\{ p \in \mathbb{C}^8 : \sum_{i \in I_f} p_i = \sum_{i \in I_g} p_i = 0 \right\}, \qquad |I_f| = |I_g| = 4. \tag{1}$$

Its smooth locus is the complement of the eight coordinate hyperplanes. The node parameters satisfy $\tau_i(p) = p_i/u_i(p)$ for nonvanishing holomorphic units. The smooth fibres are complete intersections of bidegrees $(1,1), (1,1), (2,2)$ in $\mathbb{P}^3 \times \mathbb{P}^3$. Write $H = \mathcal{O}(2,2)$ and $L = \mathcal{O}(1,-1)$ on each fibre. The actual Ricci-flat metric in $c_1(H)$ has volume $V = 160/3$, and the split

Date: November 12, 2026
This version: November 12, 2026.
2020 *Mathematics Subject Classification.* 53C07 (primary); 14J32, 32Q25, 58J35 (secondary).
Key words and phrases. Hermitian–Yang–Mills flow, conifold degeneration, Hermitian–Einstein metric, relative spectrum, harmonic moduli metric.

bundle is

$$E^{\mathrm{sp}} = L \otimes \mathbb{C}^2 \oplus L^{-1} \otimes \mathbb{C}^2.$$

The polarized construction and the uniform actual-metric input are restated in Section 2 from [W26c]. All subsequent bundle and flow estimates are proved here.

The bundle deformation parameters are two matrices $z = (A, B) \in W = \mathrm{Mat}_2(\mathbb{C}) \oplus \mathrm{Mat}_2(\mathbb{C})$, with compact action $(k_+, k_-) \cdot (A, B) = (k_+ A k_-^{-1}, k_- B k_+^{-1})$. The two harmonic extension forms have squared norms $c_+(p), c_-(p) > 0$. They are uniformly bounded above and below and converge to positive central values. On the real seven-dimensional space

$$\mathfrak{h} = \{(S_+, S_-) : S_\pm = S_\pm^*, \quad \operatorname{tr} S_+ + \operatorname{tr} S_- = 0\}$$

define the mass and energy

$$\begin{aligned} M(S) &= V(\operatorname{tr} S_+^2 + \operatorname{tr} S_-^2), \\ E_{p,z}(S) &= 2c_+ \|S_+ A - A S_-\|_{\mathrm{HS}}^2 + 2c_- \|S_- B - B S_+\|_{\mathrm{HS}}^2. \end{aligned} \tag{2}$$

Let $\mu_1 \le \cdots \le \mu_7$ be its generalized eigenvalues, including zeros. The quadratic moment is

$$\mu_{2,p}(A, B) = V^{-1}(c_+ A A^* - c_- B^* B,\ c_- B B^* - c_+ A^* A).$$

Its zero set becomes $X X^* = Y^* Y$, $Y Y^* = X^* X$ under $X = \sqrt{c_+} A$, $Y = \sqrt{c_-} B$.

Theorem 1.1. *On the family above there is an actual holomorphic rank-four bundle family $\mathcal{E}_{p,w}$ realizing the full local deformation germ of E^{sp}, with $\det \mathcal{E}_{p,w} = \mathcal{O}$ and $c_2(\mathcal{E}_{p,w}) H = 32$. After one common shrink, the following hold uniformly over every smooth p and all small bundle parameters.*

(i) *There is a real K-equivariant map $\chi_p(z) = z + O(|z|^2)$, $d\chi_p = I + O(|z|)$, such that every $z \in \mu_{2,p}^{-1}(0)$ gives an actual HE connection on $\mathcal{E}_{p,\chi_p(z)}$. These representatives cover the sufficiently small polystable objects in the stated local germ. The germ contains no stable rank-four objects.*

(ii) *At these HE connections the positive operator linearizing $h^{-1}\dot{h} = -2\mathrm{i}\Lambda F_h$ has complete small spectrum*

$$(1 - C|z|)\mu_j \le \lambda_j \le (1 + C|z|)\mu_j \quad (1 \le j \le 7), \qquad \lambda_8 \ge c > 0.$$

Its exact trace-free kernel is the compact infinitesimal stabilizer. The comparison allows changing kernel dimension and arbitrary ratios of positive eigenvalues.

(iii) *Fix an HE member, write $r = |\chi_p(z)|$, and fix any finite graph regularity order $m \ge 4$. There is a common spatial C^1 basin of smooth determinant-normalized initial metrics and a chosen C^m seven-dimensional invariant graph. Every flow in that basin selects a C^1 finite initial phase and tracks the corresponding exact graph trajectory with error $C D_* e^{-ct}$ in the logarithmic C^1 norm. Here D_* is its initial displacement from the graph. The selected graph trajectory remains in the actual equation for all positive time. Its finite metric is the pulled-back L^2 metric, its energy is twice Donaldson's functional, and its Hessian at the HE origin is relatively $1 + O(|z|)$ equivalent to $E_{p,z}$ in the same S coordinates. The preparation time $4\sigma^{-1}\log(1/r)$ is negligible compared with every positive inverse weak rate.*

(iv) *The local HE classes are parametrized by $\mathbb{C}^2/\mathsf{W}$, where $\mathsf{W} = (\mathbb{Z}/2)^2 \rtimes S_2$ acts by signs and permutation. On every quotient stratum, their actual harmonic connection metric is relatively $1 + O(|u|)$ equivalent to $4|\mathrm{d}u|^2$. For paths confined to a fixed sufficiently small local locus, its metric completion is the corresponding closed parameter locus, and its unique tangent cone at the split bundle is $(\mathbb{C}^2, 4|\mathrm{d}u|^2)/\mathsf{W}$. The rescaled distance error is $O(\rho)$ on bounded balls, uniformly in the degenerating base.*

The theorem combines the precise statements proved below. The bundle realization is Proposition 2.2. The relative spectrum and exact balancing are Theorem 6.1 and Proposition 7.1; Section 7 proves holomorphic polystability and local completeness. The dynamical claims are Theorems 9.1, 10.1, 11.1 and Corollary 10.2. The actual quotient identification, harmonic metric and distance are Theorems 12.2, 12.1 and 13.1. Their proofs use the same connections and the same compact action. The flow parameters are fixed along each trajectory; uniformity permits them to approach their limits jointly when comparing different trajectories.

The balanced normal form has two cycle parameters. Its four types have trace-free stabilizer dimensions $1, 3, 2, 7$: two distinct nonzero cycles, a repeated nonzero cycle, one nonzero cycle and two zero vertex factors, and the split bundle. All occur in the theorem. The action in (2) controls both the zero rates and the unitary quotient directions. The graph metric, however, varies metrics on a fixed bundle, whereas the harmonic metric varies its holomorphic isomorphism class.

A simple path exhibits two widely separated rates. Take $X = Y = \operatorname{diag}(h, h+h^b)$ with fixed $b > 1$. There is one exact zero, two rates $2h^{2b}V^{-1}(1+O(h))$, and four rates $8h^2V^{-1}(1+o(1))$; see Proposition 8.2. This permits an arbitrarily long weak time scale independently of every node scale. The all-time comparison uses the exact finite field, not a truncation whose error could accumulate on that scale.

1.2. **Relation to earlier work.** Local HE reduction and geometric invariant theory on a fixed compact Kähler manifold are established tools. Buchdahl–Schumacher construct equivariant local deformations and analytic GIT quotients [BS21, Theorems 5–8]; their Proposition 5 identifies the harmonic form. Toda describes polystable sheaf germs by Ext-quivers with convergent relations [Tod18, Theorem 3.2]. Delloque proves complementary correction and local quotient results for elliptic polynomial curvature equations, including HYM [Del25, Propositions 4.3 and 4.5 and Theorem 7.4]. Here the relations vanish for a geometrically realized family, although its threefold Ext-two group is nonzero. The analytic estimates must also remain uniform as the underlying metric degenerates. The dense-stable hypothesis in [BS21, Theorem 10] is absent in our rank-four neighbourhood, so that extension theorem does not provide the harmonic distance result used here.

Nonflat HYM metrics through conifold transitions were constructed by Chuan [Chu12]. Collins–Picard–Yau prove a tangent-bundle theorem with quantitative local profiles for the Fu–Li–Yau balanced backgrounds [CPY24, Theorem 1.1]. Our background is the actual Ricci-flat Kähler metric, the bundle has rank four, and the base and bundle parameters may have independent scales. The additional conclusions concern the complete relative endomorphism spectrum, the spatial metric flow and the harmonic metric on actual bundle classes.

Haiden–Katzarkov–Kontsevich–Pandit identify iterated weight filtrations and iterated-logarithmic asymptotics for finite algebra flows [HKKP23], and prove actual bundle-flow asymptotics on curves [HKKP18, Theorem 1.1]. Their order and integrable-forcing comparison in Section 4.2 of [HKKP18] applies in arbitrary compact Kähler dimension. Lam proves logarithmic and iterated-logarithmic asymptotics on a fixed smooth compact Kähler base for semistable bundles with locally free graded object and the stated rationality condition on the invariant inner product [Lam26, Theorem 8.2]. His Proposition 8.6 compares actual and reduced trajectories at bounded distance after a preliminary complex-gauge change. Our theorem instead concerns a uniform small basin around polystable HE metrics over degenerating geometry and gives decaying spatial tracking to a selected exact graph trajectory. It does not determine nonlinear decay exponents or leading coefficients of a trajectory.

Invariant manifolds for parabolic equations are classical; see, for example, [Sim94, Theorem 2.1]. Here we prove the required C^0-to-C^1 heat estimates from scalar Ricci-flat heat bounds and controlled split connection coefficients. The logarithmic metric equation depends only on its first spatial jet. These facts give a direct graph construction without assuming uniform high-order coordinate estimates at the shrinking nodes. The metric-distance inequality then proves confinement for an open spatial basin and keeps the selected graph trajectory in the original equation.

Earlier work in this series treats stationary local reduction on a fixed smooth background [W26e], finite phase reduction for trajectories satisfying its confinement hypotheses [W26f], and all-time approximation on degenerating curves with the stated spatial initial-data class [W26g]. The present proof verifies the complementary estimates, confinement and quotient geometry anew on the nonflat conifold family. Its contribution is this geometric realization and the simultaneous uniformity of the spectral, dynamical and harmonic-metric comparisons.

1.3. **Structure of the proof.** Sections 2–5 construct the bundles and the uniform analytic estimates. Sections 6–8 prove the complete relative spectrum, exact balancing, polystability classification and multiscale example. Sections 9–11 construct the spatial basin, invariant graph and phase. Sections 12–13 identify actual classes and prove the harmonic metric and distance comparisons. All constants refer to the common small domains specified in these statements.

2. The complete-intersection family and its bundles

We use the polarized conifold family of [W26c, Section 2]. The construction below retains its period coordinates and its exact polarization. In particular, the bundle parameters will be independent of the parameters which smooth the nodes.

2.1. **The compact family and the metric input.** Put $\Sigma = \mathbb{P}^1_f \times \mathbb{P}^1_g$ and let

$$Z = \mathbb{P}_\Sigma(\mathcal{O} \oplus \mathcal{O}(1,-1)) \times_\Sigma \mathbb{P}_\Sigma(\mathcal{O} \oplus \mathcal{O}(1,-1)), \qquad k_i = g + \xi_i,$$

where the projective bundles parametrize one-dimensional quotients. A general anticanonical hypersurface $Y \subset Z$ has a small contraction $Y \to X_0$ with eight ordinary double points. Four exceptional curves have each of the two numerical types

$$(1,0,0,0), \qquad (0,1,-1,-1)$$

in the divisor basis (f, g, ξ_1, ξ_2). Write I_f, I_g for the corresponding groups of nodes. The compact construction and its intersection data are given in [W26a, Appendix A]; the polarized smoothing is developed in [W26b, Section 8] and [W26c, Section 2].

Fix the resulting proper polarized family $\pi : \mathcal{X} \to \mathcal{B}$, where $\mathcal{B}$ is a ball about zero in

$$\left\{ p \in \mathbb{C}^8 : \sum_{i \in I_f} p_i = 0, \quad \sum_{i \in I_g} p_i = 0 \right\}. \tag{3}$$

Its relative polarization is denoted by $\mathcal{H}$. The relative canonical bundle is trivial, and the local node equations are

$$z_1^2 + z_2^2 + z_3^2 + z_4^2 = \tau_i(p), \qquad \tau_i(p) = p_i / u_i(p), \tag{4}$$

where each u_i is a nonvanishing holomorphic function. These identities are exact. The smooth locus is $\mathcal{B}^\circ = \{p \in \mathcal{B} : \prod_i p_i \neq 0\}$. After shrinking the ball, all the units and their inverses are bounded. We permit every path $p \to 0$ in $\mathcal{B}^\circ$; no ratio or phase condition is imposed on the τ_i.

Let ω_p be the actual Ricci-flat Kähler form in $c_1(\mathcal{H}_p)$ and let g_p be its Riemannian metric. We recall explicitly the analytic input from [W26c, Theorem 9.1 and Section 10]. There are conic radii ϱ_p, equal up to fixed constants to $(|z|^2+|\tau_i|)^{1/3}$ on the node charts and to one on a fixed exterior, and initial Kähler metrics $\widetilde{g}_p^R$ which are normalized Stenzel metrics on the inner charts. For fixed small $R>0$ and every fixed k, there are $\mu>\eta>0$ such that

$$\limsup_{\substack{p\to 0\\ p\in\mathcal{B}^\circ}} \max_{j\le k} \sup_{X_p} \varrho_p^j |\nabla^j (g_p-\widetilde{g}_p^R)|_{\widetilde{g}_p^R} \le C_k R^{\mu-\eta}. \tag{5}$$

The constants allow arbitrary independent node scales. On each fixed compact exterior, g_p converges in every spatial derivative to the conically singular central Ricci-flat metric g_0. There is also a uniform positive scalar Poincaré constant. The latter follows either from the degree-one gap in the cited Hodge theorem, by applying d to scalar eigenfunctions, or from the uniform diameter estimate proved below. These inputs concern spatial derivatives; no derivative in p is used. The order in (5) is to fix R, take $p\to 0$, and then decrease R.

2.2. **A relative complete intersection.** Let $\mathbf{P}=\mathbb{P}^3\times\mathbb{P}^3$. The globally generated classes k_1,k_2 define a morphism $q_Z: Z\to\mathbf{P}$. Writing $[s_0:s_1],[t_0:t_1]$ on Σ, its local expression is

$$(x;y)=(v_1 s;u_1 t), \qquad (x';y')=(v_2 s;u_2 t).$$

The quotient coordinates (u_i,v_i) transform with the indicated line bundles. The image is the fourfold

$$V_0=\{x_0x_1'-x_1x_0'=0,\quad y_0y_1'-y_1y_0'=0\}\subset\mathbf{P}. \tag{6}$$

Indeed, each nonzero pair of x-vectors has a common direction s, and each nonzero pair of y-vectors a common direction t. A direction is free only if both corresponding vectors vanish; both cannot be free. The remaining scalars recover the two quotients. Thus the image is irreducible of dimension four. The two equations form a regular sequence and their intersection is generically smooth. A Cohen–Macaulay complete intersection has no embedded components, so this identifies the reduced scheme image as well.

The singular locus consists of the disjoint lines

$$\ell_x=\{x=x'=0,\ [y]=[y']\}, \qquad \ell_y=\{y=y'=0,\ [x]=[x']\}.$$

Near either line one determinant equation eliminates a smooth coordinate; the other is the ordinary threefold node transverse to the line. Hence V_0 is normal. The two exceptional surfaces of q_Z have codimension two and are contracted along one ruling. Adjunction gives

$$K_{V_0}=\mathcal{O}_{V_0}(-2,-2), \qquad -K_Z=2k_1+2k_2=q_Z^*\mathcal{O}_{V_0}(2,2).$$

The proper birational map onto the normal image has $q_{Z*}\mathcal{O}_Z=\mathcal{O}_{V_0}$. Its anticanonical section ring is therefore the section ring of $\mathcal{O}_{V_0}(2,2)$, so V_0 is exactly the anticanonical model in the compact construction. The chosen X_0 is cut out in V_0 by a section of this bundle. It lifts to a section h_0 on $\mathbf{P}$, because the Koszul sequence

$$0\longrightarrow\mathcal{O}_{\mathbf{P}}\longrightarrow\mathcal{O}_{\mathbf{P}}(1,1)^{\oplus 2}\longrightarrow I_{V_0}(2,2)\longrightarrow 0$$

has no higher cohomology in its first two terms. The restriction of h_0 to either singular line has degree four and four simple zeros. These are the original eight labelled nodes.

Proposition 2.1. *After shrinking $\mathcal{B}$, the given family admits a relative closed embedding into $\mathbf{P}\times\mathcal{B}$ as a complete intersection of bidegrees $(1,1),(1,1),(2,2)$. It identifies $\mathcal{H}$ with $\mathcal{O}_{\mathbf{P}}(2,2)|_{\mathcal{X}}$. The first two equations define a proper flat family $\mathcal{V}\to\mathcal{B}$ of normal integral Gorenstein fourfolds, and $\mathcal{X}\subset\mathcal{V}$ is a relative anticanonical Cartier divisor.*

Proof. The three-equation Koszul resolution gives $H^1(X_0,\mathcal{O}) = H^2(X_0,\mathcal{O}) = 0$. For $M_{1,0} = \mathcal{O}_{X_0}(1,0)$, the noninitial terms of that resolution are

$$\mathcal{O}(0,-1)^{\oplus 2} \oplus \mathcal{O}(-1,-2), \qquad \mathcal{O}(-1,-2) \oplus \mathcal{O}(-2,-3)^{\oplus 2}, \qquad \mathcal{O}(-3,-4).$$

They are acyclic on $\mathbf{P}$. Thus $h^0(M_{1,0}) = 4$ and its higher cohomology vanishes; the same is true of $M_{2,0} = \mathcal{O}_{X_0}(0,1)$.

The family $\mathcal{X}$ is the analytic pullback of a projective family over a smooth algebraic Hilbert neighbourhood. The relative Picard space is smooth near either class because $H^2(\mathcal{O}) = 0$, and étale there because $H^1(\mathcal{O}) = 0$. After an étale neighbourhood, a section through a smooth central point and a normalized Poincaré bundle give the exact extensions $\mathcal{M}_1, \mathcal{M}_2$; see [Kle05, Theorems 2.5(2), 4.8 and Proposition 5.19]. The analytic map from $\mathcal{B}$ lifts by local inversion of the étale map. Consequently neither the period coordinates nor the functions (4) change.

Cohomology and base change extend the four generating sections of each $M_{i,0}$ to frames of $\pi_*\mathcal{M}_i$. Their evaluation maps remain surjective, by Nakayama's lemma and properness. They give a map $\iota : \mathcal{X} \to \mathbf{P} \times \mathcal{B}$. The same Koszul calculation gives, with no higher cohomology,

$$h^0(\mathcal{O}_{X_0}(1,1)) = 16 - 2 = 14, \qquad h^0(\mathcal{O}_{X_0}(2,2)) = 100 - 32 - 1 + 1 = 68. \tag{7}$$

For $(1,1)$ only the terms $\mathcal{O}(1,1)$ and $\mathcal{O}^{\oplus 2}$ have ambient cohomology. For $(2,2)$ they are $\mathcal{O}(2,2)$, $\mathcal{O}(1,1)^{\oplus 2} \oplus \mathcal{O}$, and $\mathcal{O}$. Their degree-zero complex is exact in negative total degree, which proves the assertions rather than only their Euler characteristics. The sixteen product sections span the first space in (7). This spanning persists near zero, as does very ampleness of $\mathcal{M}_1 \otimes \mathcal{M}_2$. The composite of ι with the Segre embedding is consequently a closed immersion, so ι is one as well.

If I is its relative ideal, the bundles of sections of $I(1,1)$ and $I(2,2)$ have ranks two and thirty-two. Lift the two central determinants as a frame of the former and h_0 as a section of the latter. These three equations generate I: the ideal is flat over the base, its restriction to X_0 is the central ideal, and the cokernel of the three generators vanishes there. Nakayama and properness again remove its support.

Work first on a neighbourhood $\mathcal{T}$ of the central triple in the affine coefficient space. The regular-sequence property, normal integrality of the fourfold, and the nonzerodivisor property of the last equation are open there. The universal Koszul resolutions give flatness. Pulling back by the holomorphic coefficient map $\mathcal{B} \to \mathcal{T}$ produces precisely $\mathcal{V}$ and $\mathcal{X}$, with the asserted dualizing bundles by adjunction. Finally $\mathcal{M}_1^2\mathcal{M}_2^2$ and $\mathcal{H}$ have the same central class. Their sections of the étale relative Picard space agree near zero. Their quotient is pulled back from the base and is trivial after normalization along the chosen section. This proves the exact polarization identity. □

2.3. **Cohomology and effective bundle deformations.** Put $L = \mathcal{O}_V(1,-1)$ on any fourfold fibre of $\mathcal{V}$ and set

$$\begin{aligned} E^{\mathrm{sp}} &= L \otimes \mathbb{C}^m \ \oplus \ L^{-1} \otimes \mathbb{C}^n, \\ W_{m,n} &= \operatorname{Hom}(\mathbb{C}^n, \mathbb{C}^m) \oplus \operatorname{Hom}(\mathbb{C}^m, \mathbb{C}^n), \\ G_{m,n} &= \mathrm{GL}_m(\mathbb{C}) \times \mathrm{GL}_n(\mathbb{C}). \end{aligned}$$

The superscript denotes the split bundle. The same symbols will denote its restriction to $X \subset V$.

Proposition 2.2. *The split family has an effective holomorphic miniversal bundle deformation over $\mathcal{B} \times (W_{m,n}, 0)$, with independent base and bundle parameters. Its local equivariant*

germ has action

$$(g,h)\cdot(A,B)=(gAh^{-1},hBg^{-1}). \tag{8}$$

On V, its Ext groups in degrees zero and one are $\mathfrak{gl}_m\oplus\mathfrak{gl}_n$ and $W_{m,n}$, and all higher Ext groups vanish. Restriction to X induces an equivalence of formal local bundle-deformation groupoids. On X the Ext groups in degrees two and three are respectively $W_{m,n}^*$ and $(\mathfrak{gl}_m\oplus\mathfrak{gl}_n)^*$, but all local obstruction equations vanish.

Proof. The ambient Koszul resolution for L^2 is

$$0\longrightarrow\mathcal{O}_{\mathbf{P}}(0,-4)\longrightarrow\mathcal{O}_{\mathbf{P}}(1,-3)^{\oplus2}\longrightarrow\mathcal{O}_{\mathbf{P}}(2,-2)\longrightarrow\mathcal{O}_V(2,-2)\longrightarrow0. \tag{9}$$

Its only nonzero ambient cohomology is $H^3(\mathcal{O}_{\mathbf{P}}(0,-4))=\mathbb{C}$, in Koszul degree -2. Thus $H^1(V,L^2)=\mathbb{C}$ and all other groups vanish. Interchanging the factors proves the same assertion for L^{-2}. The structure-sheaf resolution gives $H^0(V,\mathcal{O})=\mathbb{C}$ and $H^{>0}(V,\mathcal{O})=0$. These calculations hold on the singular fibres as well. The universal Koszul complexes on $\mathbf{P}\times\mathcal{T}$ have the same single nonzero cohomology terms, so the assertions commute with base change. Choosing generators of $H^3(\mathbb{P}^3,\mathcal{O}(-4))$ fixes algebraic frames e_+,e_- for the two relative H^1 line bundles.

Since E^{sp} is locally free, its global Ext groups are the cohomology of

$$F=\operatorname{End}E^{\mathrm{sp}}=\mathcal{O}\otimes(\mathfrak{gl}_m\oplus\mathfrak{gl}_n)\oplus L^2\otimes\operatorname{Hom}(\mathbb{C}^n,\mathbb{C}^m)\oplus L^{-2}\otimes\operatorname{Hom}(\mathbb{C}^m,\mathbb{C}^n).$$

This proves the ambient groups and the action (8). Serre duality on the proper Gorenstein fourfold, applied to the locally free self-dual F, gives $H^i(F\otimes K_V)=H^{4-i}(F)^*$. The sequence

$$0\longrightarrow F\otimes K_V\longrightarrow F\longrightarrow F|_X\longrightarrow0 \tag{10}$$

therefore identifies Ext in degrees zero and one under restriction and gives the stated groups in degrees two and three.

For effectivity, use the vector-bundle locus in the algebraic stack of flat coherent sheaves on the projective flat family $V_{\mathcal{T}}\to\mathcal{T}$. Its algebraicity, affine finitely presented diagonal and relative deformation theory are supplied by [Sta, Tags 09DS, 0DLY, 0DLZ, 08VW and 08WA]. The relative obstruction to a lift over a small Artin extension with kernel J is $\operatorname{Ext}^2_V(E^{\mathrm{sp}},E^{\mathrm{sp}})\otimes J=0$. This remains the obstruction group when V itself varies. One can see it directly by lifting transition matrices on an affine cover: the defect on triple overlaps is an $\operatorname{End}E^{\mathrm{sp}}$-valued Čech two-cocycle. Its vanishing class permits a correction, differences of lifts form the Ext-one torsor, and infinitesimal automorphisms are Ext-zero. Locally lifted free modules remain free.

There is an actual smooth framed atlas for this assertion. Choose k so that $E^{\mathrm{sp}}(k,k)$ is generated and has no higher cohomology, and put $N=h^0(E^{\mathrm{sp}}(k,k))$. The open relative Quot space of locally free quotients $\mathcal{O}_V(-k,-k)^N\to E$ inducing a frame of $H^0(E(k,k))$ carries a universal bundle. It is a GL_N framing atlas for the bundle stack. If K is the presentation kernel, the Ext sequence gives $\operatorname{Ext}^1(K,E^{\mathrm{sp}})=\operatorname{Ext}^2(E^{\mathrm{sp}},E^{\mathrm{sp}})=0$. The Quot obstruction theorem and finite-presentation lifting criterion therefore make the atlas smooth over $\mathcal{T}$; see [Sta, Tags 09TU, 0CZU and 02H6]. Local freeness on the total family follows from the flat-module criterion [Sta, Tag 00MH] and properness. Choose frames along the split section. The tangent sequence there is

$$0\longrightarrow\operatorname{Ext}^0(E^{\mathrm{sp}},E^{\mathrm{sp}})\longrightarrow\mathfrak{gl}_N\longrightarrow T_{Q/\mathcal{T}}\longrightarrow\operatorname{Ext}^1(E^{\mathrm{sp}},E^{\mathrm{sp}})\longrightarrow0.$$

A holomorphic transversal to the frame orbit through that section, in relative smooth coordinates, has exactly the $2mn$ parameters of $W_{m,n}$. Pullback of the universal bundle gives the required effective family before any formal convergence question arises.

To check restriction at every Artin order, suppose an ambient lift has been chosen over an Artin algebra and a specified X-lift exists over a small extension. First lift the ambient bundle, using the vanishing Ext-two. The difference of its restriction from the specified lift lies in Ext-one on X; (10) lifts this difference uniquely to Ext-one on V. Modifying by it gives the desired lift. For two such bundles the Hom restriction sequence has $H^0(\mathrm{Hom}(-X)) = H^1(\mathrm{Hom}(-X)) = 0$ at the split point. The vanishings persist through Artin thickenings, by induction on a small extension. Thus restriction also identifies all Hom spaces and isomorphisms, including their prescribed reductions. This proves the equivalence of formal groupoids.

In particular a minimal hull on X has the form $\mathbb{C}[[z_1, \dots, z_{2mn}]]/J$ with J in the square of the maximal ideal. The effective ambient family gives a map from this hull to a power-series ring in $2mn$ variables with invertible linear term. The induced endomorphism of the power-series ring is an automorphism, so $J = 0$. The nonzero Ext-two group is therefore an obstruction space with zero obstruction map. The same lifting argument, finite presentation and semicontinuity make the restriction morphism of bundle stacks étale near these points.

It remains to obtain the asserted equivariant germ rather than only a transversal. On a finite-type neighbourhood, the bundle stack is smooth over the smooth coefficient space $\mathcal{T}$, has affine diagonal, and has reductive stabilizer $G_{m,n}$ at the split point. That finite-type point is locally closed; restrict to an open in which it is closed, as in [AHR20, proof of Theorem 1.1]. The relative étale slice [dCFI26, Theorem 2.1 and Remark 2.3] gives a commuting étale roof to

$$[\,W_{m,n} \oplus T_0\mathcal{T}\,/G_{m,n}] \longrightarrow T_0\mathcal{T}.$$

The action on the second summand is trivial. The arrows toward the normal quotients are strongly étale; the arrow toward the bundle stack can be affine and hence representable. All these are effective algebraic charts.

The given split section defines $\mathcal{T} \times BG_{m,n}$ in the bundle stack. Pull back the representable étale chart along it. The branch through the marked point is analytically invertible after shrinking $\mathcal{T}$. The connected group $G_{m,n}$ fixes this discrete étale branch, so it descends and lifts the entire split section. Trivialize its torsor locally. Strong étaleness preserves its full stabilizer; its $W_{m,n}$-coordinate is therefore fixed by $G_{m,n}$. But $W_{m,n}^{G_{m,n}} = 0$, because (tI_m, I_n) has weights $1, -1$ on the two summands. Thus the lifted section is exactly the zero section. Base change of the strongly étale arrow to $W_{m,n} \oplus T_0\mathcal{T}$ and local inversion give a $G_{m,n}$-equivariant inverse germ, by uniqueness of the branch. Pull back its actual bundle and then the fixed map $\mathcal{B} \to \mathcal{T}$. Restriction to X preserves this germ by the étale equivalence just proved. □

Equivariance here is a statement about germs: for a fixed $g \in G_{m,n}$ the equality holds wherever its two local representatives are defined. It does not require a bounded parameter ball invariant under the whole complex group. We shall choose a common small ball invariant under the compact group $U(m) \times U(n)$.

2.4. **The rank-four family and its characteristic classes.** From now on $m = n = 2$. Put $D = c_1(L|_X)$ and $H = c_1(\mathcal{H}|_X)$. If a, b are the two ambient hyperplane classes, then $[X] = 2(a+b)^3$ in $\mathbf{P}$ and $\int_{\mathbf{P}} a^3b^3 = 1$. Consequently

$$a^3 = b^3 = 2, \qquad a^2b = ab^2 = 6, \qquad D = a - b, \quad H = 2a + 2b,$$

and

$$H^3 = 320, \qquad DH^2 = 0, \qquad D^2H = -16, \qquad V := \mathrm{Vol}(X_p, g_p) = \frac{160}{3}. \tag{11}$$

The rank-four family $\mathcal{E}_{p,z}$, $z=(A,B)\in W:=\operatorname{End}(\mathbb{C}^2)\oplus\operatorname{End}(\mathbb{C}^2)$, has

$$\det\mathcal{E}_{p,z}=\mathcal{O},\qquad c_2(\mathcal{E}_{p,z})=-2D^2,\qquad c_3(\mathcal{E}_{p,z})=0,\qquad c_2(\mathcal{E}_{p,z})H=32. \tag{12}$$

The Chern identities follow from $(1+D)^2(1-D)^2$ and constancy in the deformation family. For the determinant itself, $H^1(V,\mathcal{O})=0$ makes the relative Picard discrete near the trivial bundle. Its class stays trivial and its remaining base line bundle is trivialized locally. This supplies a determinant trivialization for the family.

The lines L,L^{-1} have zero slope and are distinct, since $D^2H\neq 0$. Thus E_p^{sp} is slope polystable with these two stable types and multiplicity two each. Interchanging the ambient factors in the Koszul Euler characteristic shows that the two lines have equal Hilbert polynomials. The split bundle is also Gieseker polystable, on the central fibre as well as nearby fibres. The nonzero pairing in (12) excludes a flat connection. Every sufficiently small matrix parameter describes an actual integrable bundle; its stability and its Hermitian–Einstein metrics will be determined below.

3. Uniform line metrics and the split endomorphism gap

All norms in this section use the actual Ricci-flat metric g_p. The positive scalar Laplacian is $\Delta=\mathrm{d}^*\mathrm{d}$, and $\mathrm{d}V_p=\omega_p^3/3!$. For a line metric h we use $F_h=\bar\partial(h^{-1}\partial h)$, so that $c_1(L)=[\mathrm{i}F_h/(2\pi)]$. In particular

$$\mathrm{i}\Lambda\partial\bar\partial u=-\tfrac12\Delta u,\qquad F_{h^{\mathrm{ref}}e^{-u}}=F_{h^{\mathrm{ref}}}+\partial\bar\partial u. \tag{13}$$

Let h_p^{ref} be the restriction of the product line metric on $\mathcal{O}_{\mathbf{P}}(1,-1)$.

Proposition 3.1. *There is a unique real function u_p of mean zero such that $h_p=h_p^{\mathrm{ref}}e^{-u_p}$ has $\mathrm{i}\Lambda F_{h_p}=0$. The functions u_p are uniformly bounded on $\mathcal{B}^\circ$ near zero, and have uniform bounds in every spatial derivative on each fixed compact exterior. The curvature form $\alpha_p=\mathrm{i}F_{h_p}/(2\pi)$ is the primitive harmonic representative of D, with*

$$\|\alpha_p\|_2^2=16,\qquad \|F_{h_p}\|_2^2=64\pi^2. \tag{14}$$

3.1. **Diameter and the scalar inverse.** We first establish the geometric estimates used in the proof:

$$\operatorname{diam}(X_p,g_p)\le D_*,\qquad g_p\ge c\,\iota_p^*g_{\mathbf{P}}. \tag{15}$$

Here $g_{\mathbf{P}}$ is a fixed product Fubini–Study metric. Choose one auxiliary radius R in (5) so small that its zeroth-order error is less than $1/4$. For all sufficiently small p, g_p is uniformly comparable with $\widetilde{g}_p^R$. On a node chart the normalized Stenzel metric satisfies

$$\sigma_{i,p}\ge c(|z|^2+|\tau_i|)^{-1/3}g_{\mathrm{eucl}}|_{Q(z)=\tau_i}. \tag{16}$$

To check the normalization, the metric on $Q(Z)=1$ is bounded below by $c(1+|Z|^2)^{-1/3}g_{\mathrm{eucl}}$, by positivity on the compact core and the conic expansion on its end. The change $z=\tau_i^{1/2}Z$ scales the Stenzel metric by $|\tau_i|^{2/3}$ and the Euclidean metric by $|\tau_i|$. This proves (16); phase rotations are unitary and the volume-normalizing constants stay in a compact positive interval.

The embedding ι is holomorphic on the total local hypersurface. Its coordinate functions extend holomorphically to the ambient chart, so their spatial derivatives are uniformly bounded on smaller charts. Hence $\iota_p^*g_{\mathbf{P}}\le Cg_{\mathrm{eucl}}$ there. On the fixed transition annuli and exterior the initial metrics converge smoothly to a positive metric. This proves the second inequality of (15). A Stenzel region of fixed conic radius has uniformly bounded intrinsic

diameter: its core has diameter $O(|\tau_i|^{1/3})$, and its conical end has bounded radial length and angular diameter. The connected fixed exterior has bounded intrinsic diameter by smooth convergence. Joining each cap to that exterior proves the first inequality. In particular the logarithmic length governing capacity does not produce unbounded intrinsic diameter. The ambient reference curvature consequently satisfies

$$\|\mathrm{i}\Lambda_{\omega_p} F_{h_p^{\mathrm{ref}}}\|_\infty \le C_{\mathrm{ref}}. \tag{17}$$

For completeness we derive the uniform scalar estimate used here from the Li–Yau Harnack inequality. On a connected compact real n-manifold with nonnegative Ricci curvature and diameter at most D_*, let $H_t(x,y)$ be the kernel of $e^{-t\Delta}$. The inequality [LY86, Theorem 2.3], with zero lower Ricci bound, gives

$$H_s(x,z) \le H_t(y,z)(t/s)^{n/2} \exp\left(\frac{d(x,y)^2}{4(t-s)}\right).$$

Set $T = D_*^2$, $s = T/2$, $t = T$, and integrate in x. The heat kernel has mass one, so

$$H_T(y,z) \ge \frac{c_n}{\mathrm{Vol}(X)}, \qquad c_n = 2^{-n/2} e^{-1/2}.$$

Thus $e^{-T\Delta} = c_n \Pi + (1-c_n)Q$, where Π is averaging and Q has a nonnegative mass-one kernel. For mean-zero bounded f, iteration and the L^∞ contraction at intermediate times yield

$$\Delta^{-1} f = \int_0^\infty e^{-t\Delta} f \,\mathrm{d}t, \qquad \|\Delta^{-1} f\|_\infty \le 2^{n/2}\sqrt{e}\, D_*^2 \|f\|_\infty. \tag{18}$$

The integral converges uniformly. Testing against smooth functions proves the equation, and ordinary elliptic regularity on each smooth fibre gives the classical solution. It has mean zero and is unique. No injectivity-radius or sectional-curvature lower bound enters this argument.

Proof of Proposition 3.1. Take $f_p = 2\mathrm{i}\Lambda F_{h_p^{\mathrm{ref}}}$. By (11),

$$\int_{X_p} f_p \,\mathrm{d}V_p = 2\int_{X_p} \mathrm{i}F_{h_p^{\mathrm{ref}}} \wedge \frac{\omega_p^2}{2} = 2\pi D H^2 = 0.$$

Apply (18) in real dimension six and use (17). The mean-zero solution of $\Delta_p u_p = f_p$ satisfies

$$\|u_p\|_\infty \le 16\sqrt{e}\, D_*^2 C_{\mathrm{ref}}.$$

Equation (13) proves the HE condition and uniqueness. In particular h_p and h_p^{ref} are globally comparable. On a fixed exterior the coefficients of the equation have uniform bounds in all spatial derivatives and uniform ellipticity. Interior estimates on a slightly larger exterior give the stated derivative bounds for u_p and its Chern connection. Fibrewise line-bundle existence is also [Kob87, Proposition 4.2.4]; the argument above supplies the uniform bound on this family.

The real form α_p is closed, of type $(1,1)$, and primitive. On a Kähler threefold $*\alpha_p = -\alpha_p \wedge \omega_p$, so it is coclosed and harmonic. Therefore

$$\|\alpha_p\|_2^2 = -\int_{X_p} \alpha_p^2 \wedge \omega_p = -D^2 H = 16.$$

This gives (14) with the specified curvature convention. □

3.2. **The complete split kernel and gap.** Equip $E_p^{\mathrm{sp}} = L_p \otimes \mathbb{C}^2 \oplus L_p^{-1} \otimes \mathbb{C}^2$ with $h_{0,p} = h_p \otimes I_2 \oplus h_p^{-1} \otimes I_2$. Let $D_{0,p}$ be the induced connection on endomorphisms. It will be abbreviated to D_0 on a fixed fibre. The section Bochner identity [Dem12, Chapter VII, Corollary 1.3] gives

$$D_0^* D_0 = 2\bar\partial_0^* \bar\partial_0, \tag{19}$$

because the contracted curvature acts by commutation with $\mathrm{i}\Lambda F_{h_{0,p}} = 0$.

Proposition 3.2. *The kernel of $D_0^* D_0$ on complex endomorphism sections is exactly the constant diagonal-block space $\mathfrak{gl}_2 \oplus \mathfrak{gl}_2$. There is $c > 0$, independent of all node-scale ratios, such that*

$$\|D_0 s\|_2^2 \geq c\|s\|_2^2 \quad \text{if } s \perp \ker D_0. \tag{20}$$

On trace-free Hermitian sections the kernel is

$$\mathfrak{h} = \{(S_+, S_-) \in \mathrm{Herm}_2^2 : \operatorname{tr} S_+ + \operatorname{tr} S_- = 0\}, \tag{21}$$

a real seven-dimensional space, with exact mass

$$M(S,T) = V\{\operatorname{tr}(S_+ T_+) + \operatorname{tr}(S_- T_-)\}. \tag{22}$$

Proof. The endomorphism bundle splits orthogonally into the two trivial matrix blocks and the $L^{\pm 2}$ matrix blocks. The former have the constant matrices as kernel, and the scalar Poincaré estimate controls their orthogonal complement. The latter have no holomorphic sections, by Proposition 2.2; their covariant and Dolbeault section energies agree up to the factor two in (19). We must show their gaps are uniform.

Suppose, to the contrary, that $p_j \to 0$ and sections of $L^2_{p_j}$ satisfy $\|s_j\|_2 = 1$ and $\|D_0 s_j\|_2 \to 0$. The Kato inequality and scalar Poincaré estimate imply

$$\big\| |s_j| - a_j \big\|_2 \longrightarrow 0, \qquad a_j = V^{-1} \int |s_j| \, \mathrm{d}V_{p_j} \longrightarrow V^{-1/2}. \tag{23}$$

Indeed $|\mathrm{d}|s_j|| \leq |D_0 s_j|$ almost everywhere, with regularization at the zero set, and $1 = \||s_j| - a_j\|_2^2 + V a_j^2$. Choose a connected fixed compact exterior C containing all topology outside small node neighbourhoods. The relative line bundle gives fixed smooth identifications there. Proposition 3.1 allows a subsequence whose connections and metrics converge in every spatial derivative on a slightly larger exterior. In finitely many frames, the sections are bounded in $W^{1,2}$. Rellich compactness and (23) give a smooth parallel limit s_0 on the interior of C, with constant norm $V^{-1/2}$. Thus the limiting connection of L^2 would be flat on C.

This contradicts the line bundle's fixed exterior class. Removing the exceptional curves from the small resolution Y preserves $H^2(Y;\mathbb{R})$: their real codimension is four, so the relative groups in degrees two and three in the Thom sequence vanish. The punctured resolution retracts onto C. The class of L there is $D_Y = k_1 - k_2$, and $D_Y^2 H = -16$, hence $2D_Y|_C \neq 0$. The limiting curvature still represents this class, either by its fixed reference connection and scalar correction, or by integration over fixed exterior two-cycles. It cannot vanish. The same argument applies to L^{-2}.

Failure of a uniform bound on any smaller period ball would produce the sequence just excluded, irrespective of the relative node scales. Taking the minimum of the scalar and off-diagonal gaps proves (20). The parallel Hermitian blocks each have real dimension four; one trace condition leaves seven. Their pointwise norm is the ordinary Hilbert–Schmidt norm, because the line metric cancels on endomorphisms. Integrating gives (22). □

These estimates separate two geometric roles. The curvature α_p represents the surviving divisor class D and has the fixed mass (14). The weak endomorphisms to be studied are deformations of (21). Neither space is the lost two-form cohomology quotient governing the positive small Hodge spectrum. The base metrics are shared, while the operators and their finite-dimensional spaces are different.

4. Harmonic extensions and Dolbeault compactness

The algebraic generators e_+, e_- fixed in the Koszul calculation of Section 2 determine classes in $H^1(X_p, L_p^{\pm 2})$ over the entire period ball. Write $\eta_p^\pm$ for their harmonic representatives for g_p and $h_p^{\pm 2}$. Their masses enter the finite bundle energy. We first identify these coefficients by elliptic problems on the central geometry, and then prove that no additional Dolbeault modes concentrate at the nodes.

Proposition 4.1. *There are closed representatives $\beta_p^\pm$ of $e_\pm$, supported on a fixed exterior, and sections $f_p^\pm$ such that*

$$\eta_p^\pm = \beta_p^\pm - \bar\partial f_p^\pm, \qquad \bar\partial^* \bar\partial f_p^\pm = \bar\partial^* \beta_p^\pm. \tag{24}$$

The sections have uniform global L^2 and L^∞ bounds and all fixed-exterior derivative bounds. For $\varepsilon_i \le r < r_0$, where $\varepsilon_i \asymp |\tau_i|^{1/3}$,

$$\int_{\{\varrho_{i,p} < r\}} |\eta_p^\pm|^2 \, \mathrm{d}V_p \le C r^4. \tag{25}$$

The metrics h_p, the sections $f_p^\pm$ and the forms $\eta_p^\pm$ converge in every fixed-exterior derivative through the full domain $\mathcal{B}^\circ$ to unique central limits. In particular

$$c_\pm(p) := \|\eta_p^\pm\|_2^2 \longrightarrow c_{\pm,0} := \|\eta_0^\pm\|_2^2 > 0. \tag{26}$$

The full harmonic degree-one endomorphism space is

$$z = (A, B) \longmapsto \eta_p(z) = \begin{pmatrix} 0 & \eta_p^+ A \\ \eta_p^- B & 0 \end{pmatrix}, \tag{27}$$

with Hermitian mass

$$M_p^{\mathrm{ext}}(z, z') = c_+(p) \operatorname{tr}(A^* A') + c_-(p) \operatorname{tr}(B^* B'). \tag{28}$$

Proof. Put $\mathcal{F}^\pm = \mathcal{L}^{\pm 2}$ on $\mathcal{X}$. The cohomology calculation and base change give $\pi_* \mathcal{F}^\pm = 0$ and $R^1 \pi_* \mathcal{F}^\pm \simeq \mathcal{O}_{\mathcal{B}}$. Leray therefore identifies $H^1(\mathcal{X}, \mathcal{F}^\pm)$ with the sections of the latter line bundle. The fixed frame gives an actual extension

$$0 \longrightarrow \mathcal{F}^\pm \xrightarrow{j} \mathcal{U}^\pm \xrightarrow{q} \mathcal{O}_{\mathcal{X}} \longrightarrow 0. \tag{29}$$

Its middle term is locally free because the free quotient locally lifts and splits. The extension restricts to the chosen class on every fibre, including X_0; see [Sta, Tags 01F2 and 0B3A] for these conventions.

Each total node neighbourhood can be chosen Stein, as a closed hypersurface in a Stein coordinate product. Cartan's Theorem B [Car52, exposé 18, p. 18-7; exposé 19, Section 5] gives a holomorphic lift s_i of 1 there. Choose a global smooth lift s_{gl} on the smooth total space and disjoint cutoffs χ_i supported in those neighbourhoods and equal to one on smaller fixed ones. Then

$$s = s_{\mathrm{gl}} + \sum_i \chi_i (s_i - s_{\mathrm{gl}}), \qquad \beta = j^{-1} \bar\partial s$$

is a smooth splitting and its extension form. The form is closed and vanishes on the smaller node neighbourhoods. Its vertical restrictions are the required representatives β_p. The fixed exterior is a smooth proper family, so these representatives have uniform bounds in every spatial derivative there. This uses the usual smooth splitting interpretation of extension classes [Dem12, Chapter V, Section 14]; no singular Dolbeault resolution is required.

The geometric comparison used in (15) gives radial cutoffs and volume bounds

$$|\mathrm{d}\chi_{r,p}| \le C/r, \qquad \mathrm{Vol}\{\varrho_{i,p} < 2r\} \le Cr^6, \qquad \varepsilon_i \le r < r_0. \tag{30}$$

These follow from bounded-core and conic-end estimates on the unit Stenzel metric and homothety. The same estimates hold centrally. Since u_p in Proposition 3.1 is uniformly bounded, subsequences converge on exteriors to bounded solutions of the central line equation. The normalization has mean zero in the limit: for a fixed r the exterior integrals converge, while the omitted integrals are $O(r^6)$. If two bounded normalized limits differ by w, then w is harmonic. A cutoff ζ_r vanishing at the tips and equal to one outside radius $2r$ gives

$$\int \zeta_r^2 |\mathrm{d}w|^2 \le 4 \int |\mathrm{d}\zeta_r|^2 |w|^2 \le Cr^4.$$

Letting $r \downarrow 0$ shows that w is constant, and its mean is zero. Thus the central HE line metric h_0 is unique and h_p converges without subsequences on exteriors.

On either $L_p^{\pm 2}$, the operator $P_p = \bar\partial^* \bar\partial$ has a uniform positive section gap by Proposition 3.2. Solve $P_p f_p = \bar\partial^* \beta_p$. The forcing is uniformly bounded and supported on the fixed exterior. The gap gives a uniform L^2 bound; interior elliptic estimates give all exterior derivative bounds. The form in (24) is closed and coclosed, and hence is the harmonic representative. Orthogonal projection bounds its norm by $\|\beta_p\|_2$. Inside each smaller node neighbourhood the forcing is zero, and the section Bochner identity gives $D_0^* D_0 f_p = 0$. Thus

$$\Delta |f_p|^2 = -2 |D_0 f_p|^2 \le 0.$$

The maximum principle on the smooth cap bounds $|f_p|$ by its uniformly bounded outer boundary values. This proves its global L^∞ bound. Testing the covariant harmonic equation with $\chi_{r,p}^2 f_p$ gives

$$\int \chi_{r,p}^2 |D_0 f_p|^2 \le 4 \int |\mathrm{d}\chi_{r,p}|^2 |f_p|^2 \le Cr^4.$$

Since $\eta_p = -\bar\partial f_p$ there, this proves (25).

An exterior subsequential limit f_0 is bounded and solves

$$\bar\partial_0^* \bar\partial_0 f_0 = \bar\partial_0^* \beta_0 \quad \text{on } X_0^{\mathrm{reg}}.$$

Two such solutions differ by a bounded covariantly harmonic section. The same global cutoff estimate forces their difference to be parallel. The exterior class $\pm 2D$ used in Proposition 3.2 excludes a nonzero parallel section. This proves uniqueness and full exterior convergence of f_p and η_p. Fatou's lemma gives the central version of (25). To compare norms, split at a fixed radius r: exterior integrals converge and both discarded contributions are $O(r^4)$. Taking $p \to 0$ and then $r \downarrow 0$ proves norm convergence.

If η_0 vanished, $s_0 - j f_0$ would be a holomorphic lift of 1 on X_0^{reg} in (29). Near a node s_0 is holomorphic and $\beta_0 = 0$, so f_0 is holomorphic there off the node. Normal Hartogs extension, in a local holomorphic frame, extends it across the node [Dem12, Chapter II, Corollary 7.8]. The lift then splits the extension on all of X_0, contradicting the nonzero class $e_{\pm,0}$. Thus both limiting masses are positive. Finally $H^1(\mathcal{O}) = 0$ and $h^1(L^{\pm 2}) = 1$. The parallel splitting of

the endomorphism bundle therefore gives all harmonic classes in (27). Opposite matrix blocks are pointwise orthogonal and the trace pairing factors within each block, proving (28). □

The numbers $c_{\pm,0}$ are specified by the two bounded central section problems with their algebraically normalized classes. They are geometric elliptic data, rather than intersection numbers. Changing a generator rescales its mass and inversely changes its arrow coordinate. No equality between the two masses is imposed.

4.1. **A cap estimate for the twisted Dolbeault operator.** For $F_p = L_p^{\pm 2}$ or $\mathcal{O}$, set

$$Q_{p,q}(v) = \|\bar\partial v\|_2^2 + \|\bar\partial^* v\|_2^2, \qquad v \in A^{0,q}(X_p, F_p).$$

The following estimate controls arbitrary almost-harmonic forms, beyond the representatives just constructed.

Lemma 4.2. *For fixed sufficiently small $s > 0$ and all sufficiently small p depending on s, uniformly in $0 \le q \le 3$,*

$$\|v\|^2_{\{\varrho_i < s\}} \le C s^2 Q_{p,q}(v) + C\|v\|^2_{\{s<\varrho_i<2s\}}. \tag{31}$$

The constant is independent of s, p and the node-scale ratios. The same statement holds for split endomorphism coefficients.

Proof. In the unitary frame associated with a holomorphic line frame on a total node neighbourhood, the connection form is $a_p = \frac{1}{2}(\partial - \bar\partial)\log h_p$ (multiplied by ± 2 for F_p). We first claim

$$\lim_{s\downarrow 0} \limsup_{p\to 0} \sup_{\varrho_i < s} \varrho_i |a_p| = 0. \tag{32}$$

The reference metric contributes zero, by the ambient lower bound (15). If the claim failed for the bounded potential u_p, there would be points with $r_j = \varrho_i(x_j) \to 0$ and $r_j|\mathrm{d}u_{p_j}(x_j)| \ge c > 0$. Pass to a subsequence with $\varepsilon_i/r_j \to \theta \ge 0$. The metrics rescaled by r_j^{-2} converge on compact sets to a complete Stenzel metric if $\theta > 0$, and on compact annuli to the punctured cone if $\theta = 0$. This follows from (5), first for fixed auxiliary radius and then with that radius decreasing, and the explicit scaling of the initial cores. The rescaled bounded scalar forcing tends to zero. Interior $W^{2,t}$ estimates for $t > 6$ give a bounded harmonic limit in C^1 on compact sets with nonzero derivative at the base point.

A bounded harmonic function on the complete Stenzel metric is constant: it belongs, by the AC derivative estimates, to the harmonic space of growth below three in [W26c, Lemma 6.1], whose nonconstant basis elements have positive growth and independent leading terms. On the punctured cone, expand in link scalar eigenfunctions. A coefficient has the form

$$b(r) = c_+ r^{-2+\sqrt{4+\lambda}} + c_- r^{-2-\sqrt{4+\lambda}}.$$

Boundedness at zero and infinity leaves only the constant for $\lambda = 0$. Both alternatives contradict the nonzero derivative. This proves (32).

Use the spin structure defined by the parallel canonical form on the Calabi–Yau threefold. Its complex spinors are $\Lambda^{0,*}$ and its Dirac operator is $\mathcal{D} = \sqrt{2}(\bar\partial + \bar\partial^*)$. On the Ricci-flat cone the radial unitary identification gives

$$\mathcal{D} = J\left(\partial_r + \frac{5}{2r} + \frac{A_\Sigma}{r}\right), \tag{33}$$

where J is unitary and A_Σ is the self-adjoint link spin-Dirac operator, with the appropriate chiral sign copies; see [JSZ08, Section 6, (42)] and [LM06, Sections 2.4 and 3]. The five-dimensional Einstein link has scalar curvature 20. Friedrich's inequality gives

$$|\lambda(A_\Sigma)| \geq \sqrt{\frac{5}{4(5-1)}20} = \frac{5}{2};$$

we use the primary-author formulation in [FK01]. For a compactly supported spinor put $t = \log r$ and $w = r^3\psi$. Then

$$\|\psi\|_2^2 = \int \|w\|^2 \,\mathrm{d}t, \qquad \|r\mathcal{D}\psi\|_2^2 = \int \|(\partial_t + A_\Sigma - \tfrac{1}{2})w\|^2 \,\mathrm{d}t.$$

The cross term integrates to zero. The distance of $1/2$ from the link spectrum is at least two, so

$$\|\psi\|_2 \leq \tfrac{1}{2}\|r\mathcal{D}\psi\|_2. \tag{34}$$

This is a weighted Dirac estimate, obtained before squaring the Dolbeault operator.

On the unit Stenzel cap, unit-scale interior estimates on dyadic annuli give

$$\|\rho\nabla\psi\|_2 \leq C(\|\rho\mathcal{D}\psi\|_2 + \|\psi\|_2).$$

Its differentiated AC expansion and (34) absorb the operator difference on a sufficiently distant end. A cutoff to that end yields

$$\|\psi\|_2 \leq C\|\rho\mathcal{D}\psi\|_2 + C\|\psi\|_{L^2(K)}$$

for a fixed compact core K. The complete Stenzel Dirac operator has no L^2 kernel: Lichnerowicz's formula $\mathcal{D}^2 = \nabla^*\nabla$, tested with large-radius cutoffs, makes an L^2 harmonic spinor parallel, and its infinite-volume constant norm must vanish [GL83, (2.14)]. If the compact term could not be removed, unit-norm spinors with $\rho\mathcal{D}\psi \to 0$ would have a local harmonic L^2 limit. It is zero, contradicting the displayed estimate. We obtain

$$\|\psi\|_2 + \|\rho\nabla\psi\|_2 \leq C\|\rho\mathcal{D}\psi\|_2 \tag{35}$$

for compactly supported spinors on the entire cap.

Choose the auxiliary radius in (5) comparable to, and larger than, $2s$. On $\varrho < 2s$ the initial metric is exactly Stenzel, and the scaled C^1 difference from g_p is at most $\delta(s) \to 0$ for sufficiently small p depending on s. Identify spinors by the positive orthonormal-frame map. The principal-symbol difference is $O(\delta)$ and the connection difference is $O(\delta/\varrho)$, so

$$\|\varrho(\mathcal{D}_{g_p} - \mathcal{D}_{\sigma_p})\psi\|_2 \leq C\delta(s)(\|\varrho\nabla\psi\|_2 + \|\psi\|_2).$$

The line twist adds Clifford multiplication by a_p, whose weighted norm tends to zero by (32). Homothety of (35) absorbs both errors, with constants independent of the individual cap scale. The parallel canonical form supplies the Dolbeault identification for each metric; it adds no further twist. Thus $\|\psi\|_2 \leq C\|\varrho\mathcal{D}_{F_p}\psi\|_2$ for spinors supported in the actual cap. Apply this to $\chi_s v$, with $\chi_s = 1$ inside radius s and zero outside $2s$. The commutator is Clifford multiplication by $\mathrm{d}\chi_s$, and $\varrho|\mathrm{d}\chi_s| \leq C$ on its shell. For a pure $(0,q)$-form, $\|\mathcal{D}_{F_p}v\|_2^2 = 2Q_{p,q}(v)$. Squaring proves (31). The trivial line has no connection perturbation; the orthogonal split endomorphism decomposition proves its assertion. □

4.2. **The central harmonic space and the complementary gap.** The cap estimate has the following compactness consequence. If v_j is bounded in L^2 and $Q_{p_j,q}(v_j) \to 0$, fixed-exterior Dirac estimates give local H^1 bounds. A subsequence converges strongly in local L^2 to a smooth formal harmonic form on X_0^{reg}. Fatou's lemma makes the limit L^2. For fixed s, pass to the limit in the shell term of (31); then let $s \downarrow 0$. The shell norm of the central L^2 limit tends to zero, so no mass is lost at any node. Polarization preserves finite Gram matrices. For eigenforms with eigenvalue tending to zero, the second-order equation also gives smooth exterior convergence.

To use this compactness, we identify the central formal harmonic space. The needed local primitive estimate is sensitive to the Ricci-flat density. Near a node put $s = |z|$ for the induced Euclidean radius on $Q(z) = 0$ and $r \asymp s^{2/3}$ for the cone radius. For a line-valued $(0,1)$-form the bounded line metric and central conical asymptotics give

$$g_0 \asymp s^{-2/3} g_E, \qquad \mathrm{d}V_{g_0} \asymp s^{-2}\mathrm{d}V_E, \qquad |\alpha|^2_{g_0,h_0}\mathrm{d}V_{g_0} \asymp s^{-4/3}|\alpha|^2_E \mathrm{d}V_E. \tag{36}$$

We claim that for fixed nested node neighbourhoods $U \Subset U'$ and a weakly closed $L^2(g_0,h_0)$ input, there is a primitive with

$$\bar\partial f = \alpha \quad \text{on } U^{\mathrm{reg}}, \qquad \|r^{-1}f\|_{L^2(g_0,h_0;U)} \le C\|\alpha\|_{L^2(g_0,h_0;U')}. \tag{37}$$

Here is the precise weighted application of the local Koppelman formula. In [LR18, Theorem 1.3] take $n = 3, d = 2, \nu = 1, p = 2, q = 1$. The input in (36) is Euclidean L^2. At this exponent the punctured maximal domain agrees with the whole-variety weak domain, as specified in the source's footnote 1. Its homotopy identity gives $\alpha = \bar\partial K\alpha$, with no degree-one projection term. The coefficient kernels in [LR18, (3.1)–(3.2), Lemma 3.1 and (6.4)–(6.7)], with the singular density already included, are bounded by

$$C\frac{|z|^\gamma}{|\zeta|^\gamma|\zeta - z|^5}, \qquad \gamma \in \{0,1\},$$

against $\mathrm{d}V_E$. The desired scalar output density is $s^{-10/3}|f|^2\mathrm{d}V_E$. Conjugating input by $|\zeta|^{-2/3}$ and output by $|z|^{-5/3}$ reduces the estimate to $L^2(\mathrm{d}V_E)$ boundedness of

$$T_\gamma(z,\zeta) = |z|^{\gamma-5/3}|\zeta|^{2/3-\gamma}|\zeta - z|^{-5}.$$

Use the Schur function $b(x) = |x|^{-2}$. The integral estimate [LR18, Lemma 2.5] gives

$$\int_{U'} |\zeta|^{-a}|\zeta - z|^{-5}\,\mathrm{d}V_E(\zeta) \le C|z|^{1-a}, \qquad 1 < a < 6.$$

Taking $a = 4/3 + \gamma$ in one variable and $a = 11/3 - \gamma$ in the other proves

$$\int T_\gamma(z,\zeta)b(\zeta)\,\mathrm{d}V_E(\zeta) \le Cb(z), \qquad \int T_\gamma(z,\zeta)b(z)\,\mathrm{d}V_E(z) \le Cb(\zeta).$$

All four exponents $4/3, 7/3, 11/3, 8/3$ lie strictly between one and six. The Schur test and the homotopy identity prove (37).

Let now α be a smooth formal harmonic L^2 $(0,1)$-form on X_0^{reg} with values in $F_0 = L_0^2$ or L_0^{-2}. Choose a cover with one small chart for each node and ordinary holomorphic charts elsewhere. Use (37) on the node charts and the local Dolbeault lemma on the others. The differences of the local primitives are holomorphic on overlaps. They glue the trivial extensions on these charts to a locally free extension of $\mathcal{O}_{X_0}$ by F_0, hence define a class

$$\Xi(\alpha) \in H^1(X_0, F_0). \tag{38}$$

Each node lies in just its designated chart, so the overlap functions are defined away from the nodes and satisfy the cocycle identity there. A change of local primitive on a node chart extends holomorphically by normal Hartogs; changes of primitives or cover therefore give the same extension class. Coherent GAGA identifies the analytic cohomology with the algebraically computed one-dimensional group.

The map (38) is injective. Indeed, if the extension splits, subtract its local holomorphic splittings from the primitives. They glue to f with $\bar\partial f = \alpha$ globally on the regular locus. The subtracted holomorphic functions are bounded on smaller node charts, and r^{-1} is locally square integrable in conic dimension six. Thus $f/r \in L^2$ near each node by (37). For the global cutoff ζ_r used above,

$$\bar\partial(\zeta_r f) = \zeta_r\alpha + (\bar\partial\zeta_r)f \longrightarrow \alpha \quad \text{in } L^2.$$

The second term tends to zero by absolute continuity of the integrals of $|f/r|^2$ on shrinking shells. Formal coclosedness, tested against the compactly supported $\zeta_r f$, now gives $\|\alpha\|_2^2 = 0$. By Proposition 4.1, the central space contains the nonzero form $\eta_0^{\pm}$. It consequently has dimension exactly one. This argument uses formal harmonic forms on the regular locus and does not presume an identification of central Laplacian domains.

Proposition 4.3. *The split endomorphism Dolbeault Laplacian has a uniform positive lower bound on the orthogonal complement of its harmonic kernel, in every degree $0 \le q \le 3$. Its complex harmonic dimensions in those degrees are $8, 8, 8, 8$.*

Proof. If positive degree-one eigenvalues for $L_p^{\pm 2}$ tended to zero, normalized eigenforms orthogonal to $\eta_p^{\pm}$ would, together with $\eta_p^{\pm}/\|\eta_p^{\pm}\|_2$, converge to two orthonormal central formal harmonic forms. The preceding compactness preserves both norms and their pairing. This contradicts the one-dimensional central space. The degree-zero gap is Proposition 3.2.

On each compact smooth fibre the Hodge–Serre operator is a conjugate-linear isometry from F-valued (p,q)-forms to F^*-valued $(3-p, 3-q)$-forms and satisfies

$$\Delta''_{F^*}(\#v) = \#\Delta''_F v;$$

see [Dem12, Chapter VI, Section 3.1 and proof of Theorem 7.3]. For $p = 0$ the parallel canonical form identifies its target with $(0, 3-q)$-forms valued in F^*. The dual metric of h_p^2 is h_p^{-2}. Thus the degree-one gap for the dual line gives degree two, and degree zero gives degree three, including multiplicities. The harmonic dimensions for each nontrivial line are $0, 1, 1, 0$.

For the trivial line, the scalar and degree-one gaps follow from the Hodge input and $\Delta_d = 2\Delta''$; $H^1(X_p;\mathbb{C}) = 0$. Hodge–Serre duality gives the remaining degrees. The harmonic dimensions are $1, 0, 0, 1$. Finally

$$\operatorname{End} E_p^{\mathrm{sp}} = (\mathcal{O} \otimes \operatorname{End}\mathbb{C}^2)^{\oplus 2} \oplus (L_p^2 \otimes \operatorname{End}\mathbb{C}^2) \oplus (L_p^{-2} \otimes \operatorname{End}\mathbb{C}^2)$$

is an orthogonal parallel decomposition. Summing the dimensions gives $8, 8, 8, 8$ and taking the minimum of the component gap constants proves the result. □

5. A uniform projected Hermitian–Einstein equation

We now put the effective holomorphic bundle family in controlled coordinates and solve the infinite-dimensional part of the HE equation. The estimates use the section Laplacian and first derivatives. They do not require a uniform bound for the full curvature of the degenerating background.

5.1. **The section inverse in a graph norm.** Write $P_p = \bar\partial_0^* \bar\partial_0 = \frac{1}{2} D_0^* D_0$ on complex endomorphism sections, and let $\widehat\Pi_p$ be projection onto its parallel kernel $\widehat{\mathfrak{h}} = \mathfrak{gl}_2 \oplus \mathfrak{gl}_2$. The unrestricted graph space is

$$\mathcal{G}_p = \{s \in W^{1,\infty} : P_p s \in L^\infty\}, \qquad \|s\|_{\mathcal{G}} = \|s\|_\infty + \|D_0 s\|_\infty + \|P_p s\|_\infty,$$

where the equation is distributional. On trace-free Hermitian sections write Π_p for projection onto $\mathfrak{h}$ in (21), and put

$$\mathcal{X}_p^0 = \{s \in \mathcal{G}_p : s = s^{*h_0},\ \operatorname{tr} s = 0,\ \Pi_p s = 0\}. \tag{39}$$

Lemma 5.1. *The following bounds hold uniformly on the full smooth period base near zero:*

$$\|s\|_\infty + \|D_0 s\|_\infty \le C \|P_p s\|_\infty, \qquad \widehat\Pi_p s = 0. \tag{40}$$

Consequently P_p^{-1} maps the L^∞ complement uniformly into its graph domain, and in particular maps the real trace-free complement into $\mathcal{X}_p^0$. The spaces are Banach spaces; $\mathcal{G}_p$ is a Banach algebra with a uniform product constant. Moreover $\|\eta_p^\pm\|_\infty \le C$, and harmonic projection in endomorphism degree one is uniformly bounded on L^∞.

Proof. For scalar heat flow $H_t = e^{-t\Delta}$, Bochner's formula and nonnegative Ricci curvature give

$$2t |\mathrm{d} H_t v|^2 \le H_t(v^2) - (H_t v)^2 \le \|v\|_\infty^2. \tag{41}$$

Indeed differentiation of $H_{t-a}((H_a v)^2)$ yields

$$H_t(v^2) - (H_t v)^2 = 2 \int_0^t H_{t-a}(|\mathrm{d} H_a v|^2)\, \mathrm{d}a.$$

The Bochner gradient contraction bounds each integrand below by $|\mathrm{d} H_t v|^2$. This also proves the inequality in the normalization of [Led11, Section 2, Proposition 3]. Taking real and imaginary parts and integrating $w = H_T w + \int_0^T H_t \Delta w \, \mathrm{d}t$ gives

$$\|\mathrm{d} w\|_\infty \le C T^{-1/2} \|w\|_\infty + C\sqrt{T} \|\Delta w\|_\infty. \tag{42}$$

The $t^{-1/2}$ derivative bound is integrable at zero. Applied to the bounded line potential of Proposition 3.1, this proves $\|\mathrm{d} u_p\|_\infty \le C$.

Choose a fixed finite ambient holomorphic frame cover for $\mathcal{O}(1,-1)$ and a subordinate partition with supports compactly contained in the frame charts. The restrictions of its cutoffs have uniformly bounded gradients and scalar Laplacians. The first bound follows from (15); the second follows from the Kähler trace of the fixed ambient complex Hessian. In the corresponding unitary line frame,

$$a_p = \tfrac{1}{2}(\partial - \bar\partial) \log h_p = \mathrm{i}\vartheta_p$$

is uniformly bounded. It also satisfies $\delta\vartheta_p = 0$: on a Kähler chart $\delta \mathrm{d}^c v = 0$ because the parallel complex structure contracts the symmetric real Hessian antisymmetrically. The same statements hold for the ± 2 tensor powers. For a scalar coefficient v of a section, their rough operator is

$$D_0^* D_0 v = \Delta v - 2\mathrm{i}\langle \vartheta_p, \mathrm{d} v\rangle + (\mathrm{i}\delta\vartheta_p + |\vartheta_p|^2) v. \tag{43}$$

Its first-order and zeroth-order coefficients are therefore uniformly bounded, with the divergence term zero.

Parabolic Kato and the maximum principle give

$$|e^{-tP_p} f| \le H_{t/2} |f|.$$

This follows by applying the scalar inequality to a regularized norm for the equation $\partial_t s = -\frac{1}{2}D_0^* D_0 s$; it uses no differentiated bundle domination assertion. At any fixed positive time the scalar heat kernel is uniformly bounded: apply the Li–Yau Harnack inequality to compare $H_T(x,y)$ with $H_{2T}(x,z)$ and integrate over z, using fixed volume and bounded diameter. Thus e^{-P_p} maps L^2 uniformly to L^∞. On $\widehat{\mathfrak{h}}^\perp$ the section gap gives, for $t \geq 1$,

$$\|e^{-tP_p} f\|_\infty \leq C e^{-c(t-1)} \|f\|_2 \leq C' e^{-c(t-1)} \|f\|_\infty.$$

Use contraction up to time one and integrate the semigroup to obtain $\|P_p^{-1} f\|_\infty \leq C\|f\|_\infty$.

For gradient control, localize each scalar line coefficient v_j by its ambient cutoff χ_j and extend $w_j = \chi_j v_j$ by zero to the fibre. Equation (43) gives

$$\|\Delta w_j\|_\infty \leq C\big(\|P_p s\|_\infty + \|D_0 s\|_\infty + \|s\|_\infty\big).$$

Apply (42) and sum over the fixed number of charts and matrix entries. Since the cutoffs sum to one and the connection forms are bounded,

$$\begin{aligned}\|D_0 s\|_\infty \leq C(T^{-1/2}+1)\|s\|_\infty \\ + C\sqrt{T}\big(\|P_p s\|_\infty + \|D_0 s\|_\infty + \|s\|_\infty\big).\end{aligned}$$

Choose one small T to absorb the gradient on the right. Combined with the sup-norm inverse, this is (40).

Heat-smoothing a bounded right side preserves the orthogonal complement and its sup-norm bound. On each fixed smooth fibre, the inverses have weak-star limits in $W^{1,\infty}$ which solve the equation distributionally; uniqueness identifies them with the spectral inverse. The estimates pass to the limit. The graph spaces are complete because a Cauchy sequence converges in $W^{1,\infty}$ and its P_p images converge in L^∞; distributional continuity identifies the two limits. The unconstrained kernel is $\widehat{\mathfrak{h}}$, and its intersection with the orthogonal graph domain is zero. On the full graph space,

$$P_p(uv) = (P_p u)v + u(P_p v) - \sum_\ell (D_{0,e_\ell} u)(D_{0,e_\ell} v) \tag{44}$$

proves the uniform algebra bound. The identity belongs to this unrestricted space. Product and chain rules on the weak domain follow from fixed-fibre $W^{2,q}$ regularity for every finite q; no uniform $W^{2,q}$ bound is needed.

Finally the right sides in (24) are bounded and supported on a fixed exterior. The gradient inverse gives $\|\eta_p^\pm\|_\infty \leq C$. The eight harmonic matrix basis vectors have masses bounded below by (26). Their rank-one projection formulas, with fixed volume and bounded representatives, give the L^∞ projection bound. □

5.2. **Controlled coordinates on the actual bundle family.** Let $K = U(2)\times U(2)$ act by (8). The equivariant germ in Proposition 2.2 has a representative on a common K-invariant matrix ball $|z| < a$. Indeed the étale construction has an actual group action before local inversion; uniqueness of its branch makes the local inverse and its K-translates agree, and compactness of K permits one common domain. This retains the independent full period ball after shrinking it.

We construct determinant-preserving smooth identifications of these bundles with the split bundle, holomorphic in z, which are holomorphic in the space variable on smaller fixed node neighbourhoods. The total conifold space is smooth, and the actual bundle is locally free across its central nodes. On a finite cover of a neighbourhood of the compact central fibre, choose local holomorphic bundle frames. On each node neighbourhood these give an

identification with the split bundle, equal to the identity at $z = 0$. Average it over K. The average remains holomorphic in space and parameter and is the identity at zero, hence invertible on a smaller common ball. It is K-equivariant.

Use these frames to define connections only in the matrix parameter directions. Patch them by a partition depending on the total-space point and not on z, choosing the node connection exclusively on each smaller core. Average this parameter connection over K as well. Its coefficients remain holomorphic in z. Parallel transport along $t \mapsto tz$, $0 \le t \le 1$, gives an identification T_z which is smooth in space, holomorphic in z, and equivariant. The holomorphic parameter dependence follows directly from the iterated-integral solution of its linear ODE. On the cores it agrees with the chosen holomorphic identification. The determinant trivialization can also be chosen equivariantly, with its fixed determinant character at zero, by averaging and shrinking. If $\det T_z$ differs from it by $\delta(x, z)$, then $\delta(x, 0) = 1$. Replacing T_z by $T_z\delta^{-1/4}$, with the branch near one, preserves these properties and fixes the determinant exactly.

Consequently the actual holomorphic structures become $\bar\partial_0 + \beta_p(z)$, where $\operatorname{tr}\beta_p(z) = 0$, $\beta_p(0) = 0$, and $\beta_p(z)$ vanishes on the fixed cores. Compactness of the exterior construction gives uniform spatial and matrix-parameter bounds of every fixed order. In particular

$$\|\beta_p(z)\|_\infty + \|\bar\partial_0^*\beta_p(z)\|_\infty \le C|z|. \tag{45}$$

Normalize the two scalar Kodaira–Spencer frames to the Koszul classes $e_\pm$. A required change is a nonvanishing holomorphic scalar function of p, absorbed in the arrow coordinate on a smaller common ball. It leaves the base coordinates and the two-arrow action unchanged. Write $\beta_{1,p}(z) = d_z\beta_p(0)z$. Equivariance under (tI_2, I_2) forces its diagonal blocks to vanish and its two off-diagonal blocks to have the corresponding arrow weights.

Define the linear endomorphism and complex gauge

$$f_p(z) = P_p^{-1}\bar\partial_0^*\beta_{1,p}(z), \qquad g_p(z) = e^{-f_p(z)}.$$

The inverse is on the off-diagonal blocks, whose kernel is zero. Lemma 5.1 gives $\|f_p(z)\|_{\mathcal G} \le C|z|$. It is trace-free, so the gauge has determinant one. The transformed coefficient

$$b_p(z) = g_p(z)^{-1}\beta_p(z)g_p(z) + g_p(z)^{-1}\bar\partial_0 g_p(z) \tag{46}$$

is integrable, trace-free and equivariant. Its linear part is $\beta_{1,p} - \bar\partial_0 f_p$, which is closed, coclosed and in the prescribed class, hence equals $\eta_p(z)$ in (27).

The graph algebra (44) controls the exponential remainder, including its P_p image:

$$\|g_p - I + f_p\|_{\mathcal G} + \|g_p^{-1} - I - f_p\|_{\mathcal G} \le C|z|^2.$$

For multiplication of a $(0,1)$-form by an endomorphism, the product rule for $\bar\partial_0^*$ has error bounded by the product of the endomorphism's first derivative and the form itself. Apply this on both sides of (46). The linear terms are $\bar\partial_0^*\beta_{1,p}$ and $-P_pf_p$ and cancel exactly. We have therefore proved

$$\begin{gathered} b_p(z) = \eta_p(z) + N_p(z), \qquad \|b_p(z)\|_\infty \le C|z|, \\ \|N_p(z)\|_\infty + \|\bar\partial_0^* b_p(z)\|_\infty \le C|z|^2, \qquad \|d_zN_p(z)\|_\infty \le C|z|. \end{gathered} \tag{47}$$

Holomorphic transport and the uniformly convergent exponential series give bounds for every fixed matrix derivative on a smaller common ball. The contracted derivative estimate uses P_pf_p, not a full Hessian bound for f_p.

5.3. **The projected equation.** With $b = b_p(z)$, the Chern connection for h_0 has $D_b'' = \bar\partial_0 + b$ and $D_b' = D_0' - b^*$. On endomorphisms these additional terms act by commutators. For $s \in \mathcal{X}_p^0$ define

$$\mathcal{F}_p(s,z) = e^{s/2}(\mathrm{i}\Lambda F_{h_0e^s,b})e^{-s/2}. \tag{48}$$

The conjugation makes the output h_0-Hermitian, and its trace is zero because the determinant is fixed.

Proposition 5.2. *There are uniform small radii such that, for every p and z in their domains, a unique small $s_p(z) \in \mathcal{X}_p^0$ solves*

$$(I - \Pi_p)\mathcal{F}_p(s_p(z), z) = 0. \tag{49}$$

The solution is smooth on each fibre, K-equivariant, and smooth in the real matrix parameters, with

$$\|s_p(z)\|_{\mathcal{G}} \le C|z|^2, \qquad \|d_z s_p(z)\|_{\mathcal{G}} \le C|z|. \tag{50}$$

Every fixed higher matrix derivative has a uniform bound on a smaller common ball. The remaining actual equation is

$$\Psi_p(z) := \Pi_p\mathcal{F}_p(s_p(z), z) = 0, \qquad \Psi_p(z) \in \mathfrak{h}, \quad \|\Psi_p(z)\| \le C|z|^2. \tag{51}$$

The metric $h_0e^{s_p(z)}$ is HE precisely on this zero set.

Proof. Put $M_b = \mathrm{i}\Lambda F_{h_0,b}$. Before contraction its curvature difference is

$$D_0'b - \bar\partial_0 b^* - b \wedge b^* - b^* \wedge b.$$

The contracted linear terms are controlled by $\bar\partial_0^* b$ and its adjoint, by the Kähler identities. Thus (47) gives $\|M_b\|_\infty \le C|z|^2$. The exact finite change of Chern curvature is

$$\mathrm{i}\Lambda F_{h_0e^s,b} = M_b + \mathrm{i}\Lambda\bar\partial_b(e^{-s}D_b'e^s). \tag{52}$$

This identity, with the displayed matrix order, is [CLZZ26, (2.10)]; its infinitesimal version is [Kob87, (6.3.16)–(6.3.17)]. At $(s,z) = (0,0)$ the derivative in s is

$$D_s\mathcal{F}_p(0,0)s = \mathrm{i}\Lambda\bar\partial_0 D_0's = P_p s.$$

For scalars the normalization is $\delta(\mathrm{i}\Lambda F) = \frac{1}{2}\Delta s$ for $h \mapsto he^s$.

For a general endomorphism v, expand

$$\begin{aligned}\bar\partial_b D_b'v = \bar\partial_0 D_0'v - [\bar\partial_0 b^*, v] + [b^*, \bar\partial_0 v]\\ + [b, D_0'v] - [b,[b^*,v]],\end{aligned}$$

with graded brackets for one-forms. After contraction,

$$\begin{aligned}\|\mathrm{i}\Lambda\bar\partial_b D_b'v - P_pv\|_\infty \le C\big(&\|b\|_\infty\|D_0v\|_\infty\\ &+ (\|\bar\partial_0^*b\|_\infty + \|b\|_\infty^2)\|v\|_\infty\big).\end{aligned} \tag{53}$$

No uncontracted derivative of b occurs. In (52), expand

$$\bar\partial_b(e^{-s}D_b'e^s) = (\bar\partial_b e^{-s}) \wedge D_b'e^s + e^{-s}\bar\partial_b D_b'e^s.$$

The first term uses only first derivatives. Control the second by (53) and the graph product rule. The exponential series and conjugation in (48) are uniformly smooth maps in these norms. Their weak-domain formulas are legitimate by fixed-fibre $W^{2,q}$ regularity, without any assertion of density in the graph norm.

Writing $\mathcal{F}_p(s,z) = P_p s + \mathcal{R}_p(s,z)$ gives

$$\begin{aligned} \|\mathcal{R}_p(0,z)\|_\infty &\le C|z|^2, \\ \|\mathcal{R}_p(s,z) - \mathcal{R}_p(t,z)\|_\infty &\le C(|z| + \|s\|_{\mathcal{G}} + \|t\|_{\mathcal{G}})\|s-t\|_{\mathcal{G}}. \end{aligned} \tag{54}$$

To check the vanishing factor in the second line, every term beyond $P_p s$ contains $b = O(|z|)$, an exponential remainder factor s or t, or a first derivative of one of them. The contracted derivative of b is $O(|z|^2)$. The product series and their differentiated series prove the estimate and bounded higher derivatives.

Projection Π_p is uniformly bounded on L^∞, since it is averaging against parallel block matrices with the fixed mass (22). Let J_p be the real complementary inverse from Lemma 5.1. Equation (49) is equivalent to

$$s = -J_p(I - \Pi_p)\mathcal{R}_p(s,z).$$

Choose a fixed small graph ball and then a common matrix ball so that its Lipschitz constant is less than $1/2$. The quadratic forcing makes the ball of radius $C_0|z|^2$ invariant. Contraction gives the solution and its quadratic bound. Every solution in the original small graph ball satisfies the same bound after absorption, and hence is this fixed point. Differentiating the uniformly smooth fixed-point equation gives the higher matrix bounds; $s_p(0) = d_z s_p(0) = 0$ gives the derivative estimate in (50). All operations commute with K, so uniqueness proves equivariance.

For each fixed smooth fibre the graph solution is in $W^{2,q}$ for all finite q, hence in $C^{1,\alpha}$. The coefficient of its contracted second derivative in (48) is

$$v \longmapsto e^{s/2}\left(\int_0^1 e^{-ts} v e^{ts}\,\mathrm{d}t\right) e^{-s/2}.$$

It is the identity at zero and has a smooth pointwise inverse for small s. The equation therefore expresses $P_p s$ as a $C^{0,\alpha}$ function of first derivatives and smooth fixed-fibre data. Schauder regularity and iteration make s smooth. This bootstrap uses no uniform higher unweighted spatial estimate. Finally $\Pi_p P_p s = 0$ and (54) give the size of Ψ_p. The projected equation makes the full transported curvature moment equal to Ψ_p, proving the final equivalence. □

The finite equation (51) will be solved below. At this stage the construction is an actual complementary metric correction for every small integrable matrix parameter. Its constants control matrix derivatives and the stated spatial norms throughout $\mathcal{B}^\circ$; they make no claim about derivatives in the degenerating base parameter.

6. The complete relative endomorphism spectrum

The complementary correction in Section 5 provides actual connections for every small matrix parameter, even before solving the remaining finite equation. The metric isometry $U_p(z) = e^{s_p(z)/2}$ transports the Chern connection of $(\bar\partial_0 + b_p(z), h_0 e^{s_p(z)})$ to a unitary connection $\mathcal{A}_p(z)$ for the fixed metric h_0. Its moment is exactly

$$\mathrm{i}\Lambda F_{\mathcal{A}_p(z)} = \Psi_p(z).$$

On the real Hilbert space of trace-free Hermitian endomorphisms let

$$\mathcal{L}_{p,z} = D^*_{\mathcal{A}_p(z)} D_{\mathcal{A}_p(z)}, \qquad q_{p,z}(u) = \|D_{\mathcal{A}_p(z)} u\|_2^2. \tag{55}$$

The form domain is H^1 on each compact smooth fibre; the operator has its usual elliptic domain. On the parallel space $\mathfrak{h}$ define

$$E_{p,z}(S) = 2c_+(p)\|S_+A - AS_-\|_{\mathrm{HS}}^2 + 2c_-(p)\|S_-B - BS_+\|_{\mathrm{HS}}^2. \tag{56}$$

Its mass is M in (22). This form uses all Hermitian matrix directions, including noncommuting ones. Denote its generalized eigenvalues, including zeros, by $\mu_1 \le \cdots \le \mu_7$.

Theorem 6.1. *Uniformly for $p \in \mathcal{B}^\circ$ near zero and z in a fixed small matrix ball, the whole form satisfies*

$$(1 - C|z|)\{E_{p,z}(S) + q_{p,0}(w)\} \le q_{p,z}(S + w) \le (1 + C|z|)\{E_{p,z}(S) + q_{p,0}(w)\}, \tag{57}$$

for $S \in \mathfrak{h}$, $w \perp \mathfrak{h}$ in H^1. The exact kernel consists of the constant sections

$$\{S \in \mathfrak{h} : S_+A = AS_-, \quad S_-B = BS_+\}. \tag{58}$$

The ordered eigenvalues of $\mathcal{L}_{p,z}$, including zeros, satisfy

$$(1 - C|z|)\mu_j \le \lambda_j \le (1 + C|z|)\mu_j \quad (1 \le j \le 7), \qquad \lambda_8 \ge c > 0. \tag{59}$$

The split projection Π_p identifies the whole seven-dimensional spectral space with $\mathfrak{h}$, and its inverse differs from the constant inclusion by $O(|z|)$ in L^2 operator norm. All these conclusions allow arbitrary ratios among the positive μ_j and changing kernel dimension. At a zero of Ψ_p, this is the complete small spectrum of the linearization of $h^{-1}\partial_t h = -2\mathrm{i}\Lambda F_h$.

Proof. The transported $(0,1)$ coefficient is exactly

$$U_p b_p U_p^{-1} - (\bar\partial_0 U_p)U_p^{-1}.$$

The graph and parameter bounds for s_p, together with (47), give

$$\begin{aligned} &\mathcal{A}_p(z) = \mathcal{A}_{0,p} + a_{1,p}(z) + \mathcal{N}_p(z), \qquad a_{1,p}(z) = \eta_p(z) - \eta_p(z)^*, \\ &\|a_p(z)\|_\infty \le C|z|, \quad \|\mathcal{N}_p(z)\|_\infty \le C|z|^2, \quad \|d_z\mathcal{N}_p(z)\|_\infty \le C|z|, \end{aligned} \tag{60}$$

where $a_p = \mathcal{A}_p - \mathcal{A}_{0,p}$. These maps are K-equivariant. For

$$\delta_z S = (S_+A - AS_-,\ S_-B - BS_+)$$

the infinitesimal action of $\mathrm{i}S$ is $\mathrm{i}\delta_z S$. Differentiating equivariance gives

$$[\mathrm{i}S, \mathcal{N}_p(z)] = d_z\mathcal{N}_p(z)[\mathrm{i}\delta_z S].$$

The positive lower bounds for $c_\pm$ and fixed volume therefore imply the crucial directional estimate

$$\|[\mathcal{N}_p(z), S]\|_2 \le C|z|\, E_{p,z}(S)^{1/2}. \tag{61}$$

It holds also when $E_{p,z}(S) = 0$, in which case the commutator vanishes exactly. This estimate differentiates the compact action, so it does not require a holomorphic connection remainder.

The type components and the two off-diagonal blocks are orthogonal. Hence

$$\|[a_{1,p}, S]\|_2^2 = 2\|[\eta_p, S]\|_2^2 = E_{p,z}(S).$$

Moreover $a_{1,p}$ is coclosed for D_0: the $(0,1)$ component is $\bar\partial_0$-coclosed by harmonicity, the other component is its adjoint, and the remaining adjoints have the wrong bidegrees. Since S is parallel,

$$\langle D_0 w, [a_{1,p}, S]\rangle_2 = 0. \tag{62}$$

The split section gap and (60) also give

$$\|[a_p, w]\|_2 \le C|z|\|w\|_2 \le C'|z| q_{p,0}(w)^{1/2}.$$

Now expand

$$D_{\mathcal{A}_p(z)}(S + w) = D_0 w + [a_{1,p}, S] + [\mathcal{N}_p, S] + [a_p, w].$$

By (62) the squared norm of the first two terms is exactly $q_{p,0}(w) + E_{p,z}(S)$. The other two have norm at most $C|z|$ times its square root. Squaring proves (57), first on smooth sections and then on the entire form domain.

The lower bound forces $w = 0$ and $E_{p,z}(S) = 0$ for any vector in the kernel. Conversely these conditions annihilate both commutators, including the nonlinear one by (61). This proves (58). The comparison form $E_{p,z}(S) + q_{p,0}(w)$ is an orthogonal direct sum. Its first seven eigenvalues are $\mu_1, \dots, \mu_7$, because $\mu_7 \le C|z|^2$ while the complementary gap is fixed. Min–max applied to (57) proves (59), with the exact zero multiplicities already established.

On the entire small spectral space the energy is at most $C|z|^2$ times the norm. If $v = S + w$ belongs to it, the lower comparison and split gap yield $\|w\|_2 \le C|z|\|v\|_2$. Thus Π_p is injective there and hence an isomorphism onto the seven-dimensional space $\mathfrak{h}$, with the asserted inverse estimate. This is a whole-space identification, without a claim about individual nearly coincident eigenlines.

At $\Psi_p(z) = 0$ the corrected connection is HE. The section Kähler identity gives $\mathcal{L}_{p,z} = 2\bar\partial^*_{\mathcal{A}_p(z)}\bar\partial_{\mathcal{A}_p(z)}$. The infinitesimal curvature moment for the variation $h \mapsto he^v$ is $\bar\partial^*\bar\partial v$. The prescribed metric flow therefore linearizes to $\partial_t v = -\mathcal{L}_{p,z} v$. Away from that zero set the theorem concerns the rough operator (55); it does not identify a Hessian at a noncritical metric. □

A useful second reconstruction minimizes energy at a fixed split projection. For each $S \in \mathfrak{h}$, let $w(S) \perp \mathfrak{h}$ minimize $q_{p,z}(S + w)$. Complementary coercivity gives a unique linear minimizer. The expansion in the proof shows

$$|q_{p,z}(S, w)| \le C|z| E_{p,z}(S)^{1/2} q_{p,0}(w)^{1/2}.$$

Testing its equation with $w(S)$ gives

$$q_{p,0}(w(S))^{1/2} \le C|z| E_{p,z}(S)^{1/2}. \tag{63}$$

Its mass is $M(S, S) + \|w(S)\|_2^2$, and the minimized energy is relatively equivalent to $E_{p,z}$. This variational reconstruction vanishes on the exact kernel. It need not be the inverse of the spectral projection used in Theorem 6.1.

6.1. **The quadratic term of the actual finite moment.**

Proposition 6.2. *The map in (51) satisfies*

$$\Psi_p(z) = \mu_{2,p}(z) + O(|z|^3), \qquad d_z\Psi_p(z) = d_z\mu_{2,p}(z) + O(|z|^2), \tag{64}$$

where

$$\mu_{2,p}(A, B) = \frac{1}{V}\big(c_+ AA^* - c_- B^*B,\ c_- BB^* - c_+ A^*A\big). \tag{65}$$

The estimates are uniform in the full period domain.

Proof. Pair the contracted differentiated curvature terms of M_b with a parallel block matrix. The Kähler adjoint identities and integration by parts show that their projection is zero. Consequently

$$\Pi_p M_b = \Pi_p \mathrm{i}\Lambda(-b \wedge b^* - b^* \wedge b).$$

Replacing b by η_p changes this by $O(|z|^3)$. In a unitary coframe with $\omega = \mathrm{i}\sum_j \theta^j \wedge \bar\theta^j$, if $\beta = \sum_j C_j \bar\theta^j$, then

$$\mathrm{i}\Lambda(-\beta\wedge\beta^* - \beta^*\wedge\beta) = \sum_j [C_j, C_j^*].$$

The two blocks of (27) therefore give the diagonal expression

$$\begin{pmatrix} |\eta_p^+|^2 AA^* - |\eta_p^-|^2 B^*B & 0 \\ 0 & |\eta_p^-|^2 BB^* - |\eta_p^+|^2 A^*A \end{pmatrix}.$$

Integration supplies the coefficients $c_\pm(p)$ and projection divides by V. The total trace is zero, so no further scalar correction occurs. The explicit expansion of (52) also gives

$$\|\mathcal{F}_p(s,z) - M_b - P_p s\|_\infty \le C(|z| + \|s\|_{\mathcal{G}})\|s\|_{\mathcal{G}}.$$

At $s = s_p(z) = O(|z|^2)$ this is $O(|z|^3)$, and $\Pi_p P_p s = 0$. This proves the first estimate. The uniform differentiated Taylor expansions give the second estimate. □

An additive cubic error in (64) can dominate an arbitrarily small quadratic component. We therefore solve the exact equation in the next section, instead of inferring its zeros from this expansion. For the spectral form itself, replacing $c_\pm(p)$ by $c_{\pm,0}$ is already legitimate: it changes (56) by a relative factor $1 + O(e(p))$, where

$$e(p) = \max_\pm |c_\pm(p)/c_{\pm,0} - 1| \longrightarrow 0.$$

Thus the relative spectral error is $O(|z|) + O(e(p))$ with these fixed central coefficients, for every joint base and bundle path in the stated domain.

7. Exact balancing and holomorphic polystability

The finite residual is the moment map of an actual symplectic form on the matrix parameter space. We first normalize this form by an equivariant change of real coordinates. A separate correction within complex orbits then identifies the holomorphic polystable locus.

7.1. **The actual Hamiltonian structure.** On the affine space of unitary connections on (E^{sp}, h_0), with anti-Hermitian endomorphism-valued one-form tangents, put

$$\Omega_{\mathrm{conn}}(a,b) = -\int_{X_p} \mathrm{tr}(a\wedge b)\wedge\frac{\omega_p^2}{2}. \tag{66}$$

This is a constant closed real form. For the gauge action $D \mapsto kDk^{-1}$ the infinitesimal vector is $Y_\xi = -D\xi$. In the convention $\mathrm{d}H_\xi = \iota_{Y_\xi}\Omega$, its Hamiltonian is

$$H_\xi(\mathcal{A}) = -\int_{X_p} \mathrm{tr}(\xi F_{\mathcal{A}})\wedge\frac{\omega_p^2}{2}.$$

Indeed differentiation in direction a and integration by parts give $\mathrm{d}H_\xi(a) = \int \mathrm{tr}(D\xi\wedge a)\omega_p^2/2 = \Omega_{\mathrm{conn}}(Y_\xi, a)$. This is the usual connection moment map [ACG13, Section 1.1 and Lemma 1.5], on the affine space of all unitary connections.

Pull back by the actual map $w \mapsto \mathcal{A}_p(w)$. The resulting closed K-invariant form Ω_p has, for $S \in \mathfrak{h}$, Hamiltonian

$$H_{\mathrm{i}S,p}(w) = -V\,\mathrm{tr}(S\Psi_p(w)). \tag{67}$$

At zero the linear connection map is $\eta_p(v) - \eta_p(v)^*$, so the type calculation in a unitary coframe gives

$$\Omega_{0,p}(v, v') := \Omega_p(0)(v, v') = 2\,\mathrm{Im}\{c_+ \operatorname{tr}(v_A^* v_A') + c_- \operatorname{tr}(v_B^* v_B')\}. \tag{68}$$

The sign agrees with the convention that the circle action on $(\mathbb{C}, 2\mathrm{d}x \wedge \mathrm{d}y)$ has Hamiltonian $-|z|^2$. The constant form (68) is uniformly nondegenerate. The connection derivative estimates prove $\|\Omega_p(w) - \Omega_{0,p}\| \le C|w|$, with uniform higher matrix derivatives. Thus Ω_p is symplectic on one common small ball. The quadratic Hamiltonian of the same action for $\Omega_{0,p}$ is $-V \operatorname{tr}(S\mu_{2,p}(z))$, by direct differentiation of (65).

Proposition 7.1. *There is a smooth K-equivariant diffeomorphism χ_p on a common neighbourhood of zero, whose image contains a common neighbourhood, such that*

$$\Psi_p(\chi_p(z)) = \mu_{2,p}(z), \qquad \chi_p(z) = z + O(|z|^2), \quad d\chi_p(z) = I + O(|z|). \tag{69}$$

Its fixed-order higher matrix derivatives are uniformly bounded on smaller common balls. For every $z \in \mu_{2,p}^{-1}(0)$ there is an actual HE connection $\mathcal{A}_p(\chi_p(z))$, and its complete small spectrum is given by (56) and M with relative error $O(|z|)$, including its changing kernel.

Proof. Set $\sigma_p = \Omega_p - \Omega_{0,p}$. The radial homotopy one-form

$$\alpha_p(z)(v) = \int_0^1 t\,\sigma_p(tz)(z, v)\,\mathrm{d}t$$

is K-invariant, satisfies $\mathrm{d}\alpha_p = \sigma_p$, and has size $O(|z|^2)$ and derivative $O(|z|)$. For $0 \le t \le 1$ the forms $\Omega_{p,t} = \Omega_{0,p} + t\sigma_p$ are uniformly nondegenerate on a common small ball. Define $\iota_{Z_{p,t}}\Omega_{p,t} = -\alpha_p$. Then $Z_{p,t} = O(|z|^2)$, $dZ_{p,t} = O(|z|)$, and it is equivariant. Its flow $\phi_{p,t}$ exists to time one on a smaller common ball: the scalar bound $r' \le Cr^2$ keeps its trajectories inside the original ball. Differentiating the ODE gives $\phi_{p,1} = I + O(|z|^2)$ and $d\phi_{p,1} = I + O(|z|)$, with the stated higher bounds. The reverse flow shows that its image contains a common neighbourhood. Cartan's formula gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\phi_{p,t}^*\Omega_{p,t} = \phi_{p,t}^*(\sigma_p + \mathrm{d}\iota_{Z_{p,t}}\Omega_{p,t}) = 0.$$

Thus $\chi_p := \phi_{p,1}$ pulls Ω_p back to $\Omega_{0,p}$.

Equivariance identifies the differentials of $H_{\mathrm{i}S,p} \circ \chi_p$ and $-V \operatorname{tr}(S\mu_{2,p})$. Both vanish at zero, so the Hamiltonians agree exactly. The trace pairing on $\mathfrak{h}$ is nondegenerate, proving (69). The diagonal central circle has identically zero Hamiltonian and leaves no scalar ambiguity. Therefore the actual finite zero set, on the image under consideration, is exactly $\chi_p(\mu_{2,p}^{-1}(0))$.

For $w = \chi_p(z)$, differentiate equivariance along the compact action:

$$\mathrm{i}\delta_w S = d\chi_p(z)[\mathrm{i}\delta_z S].$$

The derivative estimate and the positive bounded masses imply $(1 - C|z|)E_{p,z} \le E_{p,w} \le (1 + C|z|)E_{p,z}$, including exact equality of their kernels. Combine this with Theorem 6.1. The connection is the actual Chern connection of the constructed holomorphic bundle, and its determinant and nonzero Chern pairing are unchanged. □

The map χ_p is a real symplectic normalization. It is not asserted to be holomorphic or to preserve complex G-orbits. Its equivariance and injectivity do give $K_{\chi_p(z)} = K_z$ as compact stabilizer groups.

7.2. **Balanced normal forms and stable factors.** Introduce the normalized arrows

$$q = T_p z = (X, Y) = (\sqrt{c_+(p)}A, \sqrt{c_-(p)}B). \tag{70}$$

The linear map T_p is equivariant and uniformly invertible, and the flat balanced equations are

$$XX^* = Y^*Y, \qquad YY^* = X^*X. \tag{71}$$

Singular-value decomposition makes $X = D$ diagonal nonnegative by the K action. Then $Y^*Y = YY^* = D^2$, so Y is normal and commutes with D^2. On a positive singular-value block d, Y/d is unitary and can be diagonalized by the same unitary change at both vertices. On the zero block it vanishes. Thus every balanced point has a normal form

$$X = \operatorname{diag}(d_1, d_2), \qquad Y = \operatorname{diag}(d_1 e^{\mathrm{i}\theta_1}, d_2 e^{\mathrm{i}\theta_2}), \qquad d_j \geq 0. \tag{72}$$

Let m_λ count the entries with a given nonzero $\lambda = d_j^2 e^{\mathrm{i}\theta_j}$ and let m_0 count the zero entries. The compact stabilizer is

$$\prod_{\lambda \neq 0} U(m_\lambda) \times U(m_0) \times U(m_0). \tag{73}$$

Indeed the first stabilizer equation preserves D^2, and the second separates the phase eigenspaces on each positive block. The two zero vertex spaces transform independently.

At the actual HE connection with coordinate $\chi_p(z)$, Theorem 6.1 identifies the full Hermitian parallel algebra, after adding the scalar identity, with the stabilizer algebra in (73). Its complexification is

$$\bigoplus_{\lambda \neq 0} \operatorname{Mat}_{m_\lambda}(\mathbb{C}) \oplus \operatorname{Mat}_{m_0}(\mathbb{C}) \oplus \operatorname{Mat}_{m_0}(\mathbb{C}). \tag{74}$$

Every complex parallel section is the sum of Hermitian real and imaginary parts, so this is the entire parallel algebra. Its minimal orthogonal projections define actual parallel holomorphic subbundles. A minimal projection in a nonzero block has smooth bundle $L_p \oplus L_p^{-1}$ and only scalar parallel endomorphisms. The HE decomposition theorem [Kob87, Theorem 5.8.3 and Proposition 5.7.6] makes it slope stable, with stability understood against coherent subsheaves. It has rank two, determinant $\mathcal{O}$ and $c_2 H = 16$. Here the determinant is holomorphically trivial because its integral first Chern class is zero and $H^1(X_p, \mathcal{O}) = 0$.

Parallel matrix units identify the m_λ copies. Factors belonging to different nonzero blocks are nonisomorphic within this bundle: a holomorphic homomorphism of same-slope HE factors is parallel by the section Bochner identity on their Hom bundle. The zero blocks give L_p and L_p^{-1}, each repeated m_0 times; their holomorphic type is again fixed by injectivity of the first Chern-class map on the Picard group. Consequently the complete possibilities are

Normalized cycle type	Zero modes	Positive small modes
Two distinct nonzero cycles	1	6
One repeated nonzero cycle	3	4
One nonzero cycle and one zero entry	2	5
Both entries zero	7	0

Zero modes here are trace-free Hermitian endomorphisms. Their number is $\sum_{\lambda \neq 0} m_\lambda^2 + 2m_0^2 - 1$. All remaining eigenvalues have the complementary gap already proved. The table imposes no comparability condition on its positive small eigenvalues.

7.3. **A correction within complex orbits.** We give the finite correction from [W26d, Section 2] with the normalization and bounds needed here. For a finite quiver of total vertex dimension m, put

$$(C_qU)_a = U_{h(a)}q_a - q_aU_{t(a)}, \qquad \mathfrak{a} = \bigoplus_i \mathrm{Herm}(W_i), \qquad \omega_0(v,v') = -\operatorname{Im}\sum_a \operatorname{tr}(v_a^*v_a').$$

The incoming-minus-outgoing moment μ satisfies

$$\mathrm{d}\operatorname{tr}(\mu U) = 2\omega_0(\mathrm{i}C_qU,\cdot), \qquad \omega_0(\mathrm{i}C_qU, C_qU) = \|C_qU\|^2.$$

Suppose Ω is a closed K-invariant form on a ball B_R and ρ is its equivariant moment in this convention, normalized at zero, with

$$\rho(0) = 0, \qquad \mathrm{d}\operatorname{tr}(\rho U) = 2\Omega(\mathrm{i}C_qU,\cdot), \qquad \|\Omega - \omega_0\|_{B_R} \le \epsilon. \tag{75}$$

No type-$(1,1)$ assumption on Ω is needed.

Lemma 7.2. *Let $A_0 = 2m$ and $\epsilon < \log 2/(16A_0)$. If $x \in B_{R/4}$ satisfies $\mu(x) = 0$, there is a Hermitian tuple S such that $\rho(e^Sx) = 0$ and $\|S\|_{\mathrm{op}} \le A_0\epsilon$. It can be chosen in the centralizer of K_x. The reverse conclusion holds starting from $\rho(y) = 0$. In a fixed complex orbit, either zero set contains at most one compact orbit. These corrections induce inverse continuous maps between the two local compact quotients on nested neighbourhoods, including stabilizer changes.*

Proof. For a tuple of orthogonal projections P, split $q = n + u$ into preserving and crossing arrow blocks. The circle generated by $\mathrm{i}P$ fixes n. Both normalized paired moments vanish on its fixed linear subspace, by their Hamiltonian identities and their value at zero. Integrate along the orthogonal segment $n + tu$ to obtain

$$|\operatorname{tr}((\rho-\mu)(q)P)| \le 2\int_0^1 t\epsilon\|u\|^2\,\mathrm{d}t = \epsilon\|C_qP\|^2. \tag{76}$$

This estimate retains the crossing energy even when it is arbitrarily small.

Let $H = K_x$, let $\mathfrak{h}_x = \ker(C_x|_{\mathfrak{a}})$, and let $\mathfrak{c}$ be the Hermitian centralizer of H. Set $\mathfrak{p} = \mathfrak{c} \cap \mathfrak{h}_x^{\perp}$ for the trace pairing. On the fixed space W^H, both moments take values in $\mathfrak{p}$: equivariance gives $\mathfrak{c}$, while the generating fields for $\mathfrak{h}_x$ vanish there, so their paired moments vanish by normalization. All the homotopies below therefore have values in this fixed real vector space.

For $S \in \mathfrak{p}$ of operator norm $a > 0$, its total trace is zero. Among its m eigenvalues over all vertices there is an adjacent gap at least $a/(m-1)$; the case $m = 1$ is trivial. Let P project above such a gap. Then $\|C_xP\| > 0$. Otherwise $P, I - P \in \mathfrak{h}_x$, as they also commute with H, and orthogonality gives $\operatorname{tr}(SP) = \operatorname{tr}(S(I-P)) = 0$. The mean eigenvalues above and below a strict gap cannot both be zero.

Take $a = A_0\epsilon$ and consider on the operator-norm ball in $\mathfrak{p}$ the homotopy

$$F_t(S) = \mu(e^Sx) + t(\rho-\mu)(e^Sx), \qquad 0 \le t \le 1.$$

Every argument remains in $B_{R/2}$. At a boundary point use the projection just chosen. An upward crossing entry with eigenvalue difference $d \ge a/(m-1)$ contributes $e^{2d} - 1$ times its squared norm to the paired moment difference; a downward one contributes $1 - e^{2d}$. Since $|d| \le 2a$,

$$\begin{aligned} \operatorname{tr}((\mu(e^Sx) - \mu(x))P) &\ge 2e^{-4a}\frac{a}{m-1}\|C_xP\|^2, \\ |\operatorname{tr}((\rho-\mu)(e^Sx)P)| &\le \epsilon e^{4a}\|C_xP\|^2. \end{aligned} \tag{77}$$

Our constants make the first coefficient larger, so there is no boundary zero throughout the homotopy. At $t = 0$ the unique zero is $S = 0$, because

$$\frac{\mathrm{d}}{\mathrm{d}t}\operatorname{tr}(\mu(e^{tS}x)S) = 2\|C_{e^{tS}x}S\|^2 > 0$$

for nonzero $S \in \mathfrak{p}$. Its derivative at zero has positive quadratic form $2\|C_xS\|^2$, hence degree $+1$. Brouwer degree gives a zero at $t = 1$. If $\mathfrak{p} = 0$ or $\epsilon = 0$, the asserted zero requires no correction.

For the reverse direction form the corresponding space at y and use

$$G_t(S) = \mu(e^S y) - \mu(y) - t(\rho - \mu)(y).$$

The initial degree calculation holds without assuming $\mu(y) = 0$. The first inequality in (77) dominates (76) at y. At $t = 1$ a zero satisfies $\mu(e^S y) = \rho(y) = 0$.

For uniqueness, polar-decompose a complex transformation as ke^U and use equivariance to compare y and $e^U y$. The squared norm of $e^{tU}y$ is a sum of nonnegative exponentials and is convex. If its endpoints are in B_R, the entire path stays there. Along it,

$$\frac{\mathrm{d}}{\mathrm{d}t}\operatorname{tr}(\rho(e^{tU}y)U) \geq 2(1-\epsilon)\|C_{e^{tU}y}U\|^2.$$

If both endpoints are zeros, their paired values agree, so $C_yU = 0$ and they are compact-equivalent. The same argument with $\epsilon = 0$ proves flat uniqueness.

Finally a convergent sequence of starting representatives has bounded correction tuples. A convergent subsequence of those tuples produces a zero in the same complex orbit as the limiting starting point. Uniqueness determines its compact orbit independently of the subsequence. This proves continuity even if the centralizer space changes in the limit. With the corrections from $B_{R/4}$ to $B_{R/2}$, restriction makes the quotient maps inverse on nested neighbourhoods. More explicitly the flat quotient of $B_{R/8}$ maps onto its inverse image under the reverse map on the perturbed quotient of $B_{R/4}$, and this image contains the perturbed quotient of $B_{R/16}$. □

In our coordinates (70), set

$$\rho_p(q) = V\Psi_p(T_p^{-1}q), \qquad \widetilde{\Omega}_p = -\tfrac{1}{2}(T_p^{-1})^*\Omega_p. \tag{78}$$

Equations (67) and (68) give exactly (75), with $\widetilde{\Omega}_p(0) = \omega_0$ and $\|\widetilde{\Omega}_p - \omega_0\|_{B_R} \leq CR$. Here $m = 4$ and $A_0 = 8$. Apply the lemma with $R = 8|q_0|$, after choosing a common outer domain. A flat-balanced q_0 has an actual zero $q_1 = e^S q_0$ in the same complex orbit with

$$\|S\|_{\mathrm{op}} \leq C|q_0|, \qquad |q_1| \leq e^{2C|q_0|}|q_0|. \tag{79}$$

The origin has zero correction. The reverse assertion holds with the same type of bound. These are distinct maps from the real diffeomorphism χ_p.

7.4. **Actual equivariance and semistability.** We record how these finite transformations act on the holomorphic bundles. The effective étale roof of Proposition 2.2 has a jointly analytic group action near the identity and the zero section. After two shrinks of the base and matrix neighbourhood, compactness of the inner closure gives a group neighbourhood U_G of the identity such that

$$\mathcal{E}_{p,gw} \simeq \mathcal{E}_{p,w} \tag{80}$$

whenever $g \in U_G$ and the two representatives are in the inner region. The lifts agree by uniqueness of the inverse étale branch. If a piecewise smooth path $g(t)$ in G stays in this

coordinate region when applied to w, subdivide its compact time interval so that successive group increments lie in U_G, and compose these actual isomorphisms. The resulting isomorphism can have arbitrarily large norm. A path on $0 < t \leq 1$ is treated on every $[t, 1]$ separately, without extending the group operation to $t = 0$.

The étale chart also identifies infinitesimal automorphisms throughout a smaller neighbourhood:

$$\operatorname{Lie}\operatorname{Aut}(\mathcal{E}_{p,w}) \simeq \operatorname{Lie} G_w, \qquad \dim_{\mathbb{C}} \operatorname{Aut}(\mathcal{E}_{p,w}) = \dim_{\mathbb{C}} G_w. \tag{81}$$

Infinitesimal isomorphisms lift uniquely through the étale diagonal; see [Sta, Tag 0CJ1]. Both groups are the open unit groups in finite-dimensional complex endomorphism algebras, hence are smooth. This proves the dimension assertion without presuming that all isomorphisms between different chart points come from G.

The algebraic family before pullback is locally free and flat over its coefficient and bundle parameters. Its central split bundle is Gieseker polystable for the symmetric polarization. Indeed each line is stable as a rank-one torsion-free sheaf on the integral central fibre: a proper same-rank subsheaf subtracts a nonzero torsion Hilbert polynomial, and a line has no nonzero rank-zero subsheaf. The two reduced Hilbert polynomials coincide by the Koszul calculation. Openness of Gieseker semistability in a projective flat family [HL97, Proposition 2.3.1] therefore gives a product neighbourhood of $(0, 0)$ in which every bundle is semistable. That source does not require smooth fibres. Analytification and pullback to the prescribed period ball preserve this open condition. On smooth fibres Gieseker semistability implies slope semistability [HL97, Lemma 1.2.13].

We shall also use that a slope-polystable, Gieseker-semistable bundle is Gieseker polystable. Its slope-stable summands are locally free and Gieseker stable. Their reduced Hilbert polynomials are at most that of the whole bundle, whereas their positive rank-weighted average equals it. For all sufficiently large integers the differences are nonnegative and their weighted sum is zero. Each difference therefore vanishes as a polynomial, which proves the assertion.

Theorem 7.3. *There is a common matrix neighbourhood on the full small smooth period base in which every $\mathcal{E}_{p,w}$ is slope semistable. It is slope polystable if and only if w is a semisimple two-arrow representation, equivalently its affine complex orbit is closed. Every such object has an actual HE representative in the family constructed above. Its stable factors are the rank-two factors and line factors already described, so no bundle in this rank-four neighbourhood is slope stable. The chart contains an open neighbourhood of the split object in the local bundle stack, and the assertion applies to every polystable object in this neighbourhood.*

Proof. For a simple two-arrow representation (X, Y), the endomorphism (XY, YX) commutes with both arrows and hence is a scalar λ by Schur's lemma. If $\lambda \neq 0$, the arrows are inverse isomorphisms up to that scalar. Identifying the two spaces by one arrow makes the other λI, so simplicity forces vertex dimensions $(1, 1)$. If $\lambda = 0$ and $X \neq 0$, then $(\operatorname{im} X, 0)$ is a proper nonzero subrepresentation, since $YX = 0$; the same holds for $Y \neq 0$. With both arrows zero, a simple representation is one-dimensional at just one vertex. Thus the simple types are

$$(1, 0), \qquad (0, 1), \qquad (1, 1) \text{ with cycle } \lambda \neq 0.$$

Take any composition series and orthogonally split its flags at the vertices. Each arrow sends block j only to blocks $i \leq j$. Assign integer weight $\ell - j$ to the jth block of a length-ℓ series. The corresponding one-parameter group multiplies an i, j arrow block by t^{j-i}.

Consequently

$$q(t) = g(t)q \longrightarrow q_{\mathrm{ss}}, \qquad |q(t)|^2 = \sum_{i\le j} |t|^{2(j-i)}\|q_{ij}\|^2 \le |q|^2 \quad (|t|\le 1). \tag{82}$$

The disc is polynomial, and its limit is the direct sum of simple quotients. Orthogonal splitting ensures the norm bound in the original metric.

Balance each nonzero $(1,1)$ pair (x_j, y_j) in q_{ss} by the positive ratio $r_j = \sqrt{|y_j|/|x_j|}$ between vertex scalars. Both resulting magnitudes are $\sqrt{|x_j y_j|}$, and $2|x_j y_j| \le |x_j|^2 + |y_j|^2$. Along the Hermitian exponential path the squared norm is convex, hence never exceeds its endpoint maximum. After a unitary change the result has (72). If q is semisimple, this balanced point q_0 lies in its orbit and has $|q_0| \le |q|$. To realize a bounded path directly from q, choose $gq = q_0$ and write $g = ke^U$. The exponential path stays in the ball by convexity, and a compact-group path reaches q_0 at unchanged norm. Equation (80) and the small correction (79) give an actual HE bundle isomorphic to $\mathcal{E}_{p,T_p^{-1}q}$. This proves polystability for semisimple parameters.

For arbitrary q, (82) pulls back the actual locally free family to a holomorphic disc on the fixed smooth fibre. Each nonzero member is isomorphic to the original bundle, by the bounded-path realization on compact subintervals. The central member is the semisimple limit, which has an HE representative by the preceding argument. Choose the initial neighbourhood with slack for the final small correction, so all these parameters stay in the common Gieseker-semistable region.

The projective moduli space of Gieseker-semistable sheaves classifies S-equivalence [HL97, Theorems 4.3.3–4.3.4]. The analytified coarse moduli map composed with this holomorphic disc is constant away from zero and continuous into a separated space, so it has the same value at zero. Thus the original bundle and its semisimple limit are S-equivalent. If the original bundle were slope polystable, both would be Gieseker polystable by the preceding polynomial argument. Uniqueness of the polystable representative [HL97, Proposition 1.5.2 and Definitions 1.5.3–1.5.4] would make them isomorphic. If q were nonsemisimple, q_{ss} would lie in the boundary of its orbit. An algebraic orbit is open in its irreducible closure; every boundary orbit has smaller dimension. Hence $\dim G_{q_{\mathrm{ss}}} > \dim G_q$, contradicting (81) for the isomorphic bundles. Therefore polystability forces semisimplicity. Closed affine orbits are exactly semisimple representations by [Kin94, Proposition 3.2(i), $\theta = 0$].

For the precise factor types in these complex-orbit coordinates, choose the correction S in the Hermitian centralizer of $H = K_{q_0}$, as in Lemma 7.2. Then $H \subset K_{e^S q_0}$. The complex stabilizer of the normal form is $\prod_{\lambda\neq 0} \mathrm{GL}(m_\lambda, \mathbb{C}) \times \mathrm{GL}(m_0, \mathbb{C})^2$, whose complex dimension equals $\dim_{\mathbb{R}} H$. The stabilizer of $e^S q_0$ is conjugate to it, and the real dimension of a compact matrix subgroup cannot exceed that complex dimension: its real Lie generators are complex-linearly independent, since a Hermitian matrix cannot equal a nonzero anti-Hermitian matrix. The inclusion therefore gives equality of the compact Lie algebras. By the exact kernel theorem, the actual parallel algebra is precisely (74). Its minimal projections give the same stable rank-two factors and the two line types, with the same multiplicities. Each rank-four case has more than one stable summand. A stable bundle would be polystable and hence in one of these cases, which is impossible.

Finally the effective chart is étale as a morphism of local stacks, and its atlas is smooth and open. The image of a small analytic product neighbourhood therefore contains an open neighbourhood of the split object. Every polystable object in that image on a smooth fibre

has a representative in the chart; the criterion just proved and the bounded correction give its controlled HE representative. This is local surjectivity of the actual deformation germ. □

The controlled representative is uniformly small. A gauge taking an arbitrary unbalanced representative to it can be arbitrarily ill-conditioned. Injectivity on actual bundle isomorphism classes and the harmonic metric on their quotient will be proved in Section 12; neither follows from local surjectivity alone.

8. Moving matrix parameters and separated spectral scales

The relative comparison in Theorem 6.1 does not require fixed eigenspaces or fixed rank. In particular, cancellations between moving arrow matrices can produce scales smaller than the squared size of either arrow. We first give a finite coefficient procedure that detects these scales, and then compute an example on the actual HE locus.

8.1. **Characteristic coefficients.** Write the finite energy as $E = d^*Wd$, where d is the commutator map and $W > 0$ contains the harmonic masses. Regard the matrix-valued target as a real inner-product space with pairing $\operatorname{Re}\operatorname{tr}(U^*V)$; the adjoint d^* is taken over these real spaces. In fixed real coordinates on $\mathfrak{h}$, let $M > 0$ be its mass matrix. Define D_k by

$$\frac{\det(M + tE)}{\det M} = \sum_{k=0}^{7} D_k t^k, \qquad D_0 = 1. \tag{83}$$

Writing d in real coordinates on both domain and target, set $C = W^{1/2} d M^{-1/2}$. Cauchy–Binet gives

$$D_k = \sum_{|I|=|J|=k} |\det C_{I,J}|^2. \tag{84}$$

Thus D_k is the kth elementary symmetric function of the positive generalized eigenvalues. In this subsection order them decreasingly, as $\gamma_1 \geq \cdots \geq \gamma_\ell > 0$, where $\ell = \operatorname{rank} d$. Then $D_k > 0$ exactly for $k \leq \ell$. Writing $P_k = \gamma_1 \cdots \gamma_k$, we have $P_k \leq D_k \leq \binom{\ell}{k} P_k$, and consequently

$$\frac{\gamma_k}{\binom{\ell}{k-1}} \leq \frac{D_k}{D_{k-1}} \leq \binom{\ell}{k} \gamma_k. \tag{85}$$

These bounds identify orders of magnitude. The following statement also identifies the leading coefficients.

Proposition 8.1. *On a constant-rank path, choose $0 = m_0 < m_1 < \cdots < m_q = \ell$ and positive scales $s_1, \ldots, s_q$ with $s_{j+1}/s_j \to 0$. For each j, put $m = m_{j-1}$ and $a = m_j - m$. Suppose*

$$\frac{D_{m+k}}{D_m s_j^k} \longrightarrow c_{j,k} < \infty \quad (1 \leq k \leq a), \qquad c_{j,1} > 0, \quad c_{j,a} > 0, \tag{86}$$

and, if $m + a < \ell$, suppose also that $D_{m+a+1}/(D_m s_j^{a+1}) \to 0$. Then the a eigenvalues in the jth group, divided by s_j, converge with multiplicity to the positive roots of

$$x^a - c_{j,1} x^{a-1} + c_{j,2} x^{a-2} - \cdots + (-1)^a c_{j,a}. \tag{87}$$

The remaining $7 - \ell$ eigenvalues are exactly zero. When the matrix parameter tends to zero, these conclusions transfer to the actual small spectrum, with the same leading coefficients, by Theorem 6.1.

Proof. For the first group, (85) and the first limit bound every γ_i/s_1. Every subsequential limit of this ordered list has exactly a positive entries: its ath elementary symmetric function is positive, and its $(a+1)$st is zero when there are further entries. The positive entries have elementary symmetric functions $c_{1,k}$, so are precisely the roots of (87). Their ordered multiset is unique, which proves convergence.

Suppose the first m eigenvalues have been identified. Their smallest is much larger than the current scale s_j. Equation (85) makes γ_{m+1} comparable to s_j and bounds all later eigenvalues by a constant times s_j. For the slow list $\gamma' = (\gamma_{m+1}, \dots, \gamma_\ell)$, expansion of the elementary symmetric functions yields

$$\frac{D_m}{P_m} \longrightarrow 1, \qquad \frac{D_{m+k}}{P_m e_k(\gamma')} \longrightarrow 1 \quad (1 \le k \le \ell - m). \tag{88}$$

Indeed the term containing all m fast variables is $P_m e_k(\gamma')$. A term omitting $b \ge 1$ fast variables contains $k+b$ slow ones. The inequalities

$$e_{k+b}(\gamma') \le C_\ell \gamma_{m+1}^b e_k(\gamma'), \qquad \frac{e_{m-b}(\gamma_1, \dots, \gamma_m)}{P_m} \le C_\ell \gamma_m^{-b}$$

bound its relative contribution by $C_\ell(\gamma_{m+1}/\gamma_m)^b$. Summing proves (88), even when $e_k(\gamma')$ is arbitrarily small. The same argument gives the first formula with $k = 0$. The hypotheses now give the elementary symmetric limits of the remaining eigenvalues divided by s_j. The first step identifies exactly a positive limiting entries and all later entries as zero. Induction proves the assertion. Rank determines the exact kernel, and relative comparison transfers each positive asymptotic without dividing an absolute error by its scale. □

Every sequence admits a subsequence with constant rank and a hierarchy of this kind. To see this, take limits of all adjacent ratios $(D_{k+1}/D_k)/(D_k/D_{k-1})$, which are bounded above by (85). Place a break where the limit is zero. Within each group the eigenvalue ratios are bounded above and below. Normalize by its first coefficient ratio and pass to a further subsequence for the normalized eigenvalues. Equation (88) then verifies the required coefficient limits. This is a subsequential assertion; an oscillating path need not have unique leading coefficients.

If $D_k \sim a_k h^{\nu_k}$ with $a_k > 0$, then the positive spectral exponents are $\alpha_k = \nu_k - \nu_{k-1}$, where $\nu_0 = 0$. They are nondecreasing. Equal consecutive exponents form a group, and its polynomial coefficients are $c_{j,k} = a_{m+k}/a_m$. Formula (84) requires cancellations to be computed inside each minor; the sum of squares has no further sign cancellation. Analytic maps and a nondegenerate analytic mass allow these orders to be read from their Taylor expansions. If only $M \to M_0 > 0$ is known, one may replace M by M_0 for leading spectral and characteristic-coefficient asymptotics by relative mass comparison. This does not identify the order of each individual normalized minor.

8.2. **Two scales on actual HE bundles.** In the normalized arrows of Section 7, take $X = Y = D = \operatorname{diag}(d, e)$ with $d, e \ge 0$. Write $z = T_p^{-1}(D, D)$ and $w = \chi_p(z)$. The actual connection $\mathcal{A}_p(w)$ is HE, and its finite energy and mass are

$$\begin{aligned} E(S,T) &= 2\{\|SD - DT\|_{\mathrm{HS}}^2 + \|TD - DS\|_{\mathrm{HS}}^2\}, \\ M(S,T) &= V\{\operatorname{tr} S^2 + \operatorname{tr} T^2\}, \qquad \operatorname{tr} S + \operatorname{tr} T = 0, \end{aligned} \tag{89}$$

for Hermitian S, T.

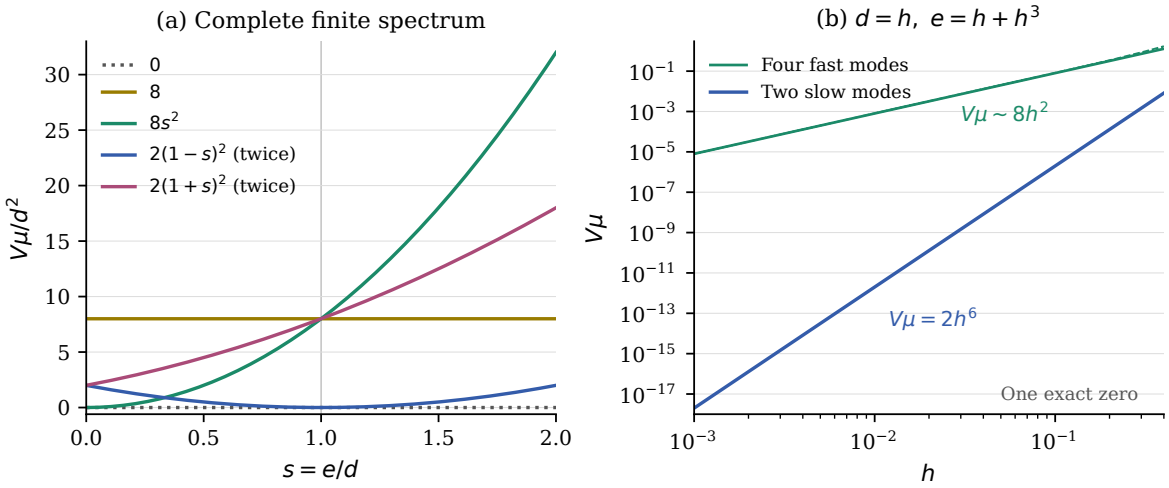


FIGURE 1. The exact finite spectrum in (90). Panel (a) shows all seven eigenvalues, with the two double branches labelled. At $e = d > 0$ the kernel has dimension three. Panel (b) takes $d = h$, $e = h + h^3$: two modes have scale h^6, four have scale h^2, and one is exactly zero. These are finite matrix calculations. The relative theorem transfers their asymptotics to the actual HYM linearization.

Proposition 8.2. *The unordered generalized spectrum of* (89) *is*

$$0, \quad \frac{8d^2}{V}, \quad \frac{8e^2}{V}, \quad \frac{2(d-e)^2}{V} \text{ (twice)}, \quad \frac{2(d+e)^2}{V} \text{ (twice)}. \tag{90}$$

For $d = h$, $e = h + h^b$, with fixed $b > 1$, the actual HYM linearization has one exact trace-free zero, two positive eigenvalues $2h^{2b}V^{-1}(1+O(h))$, four positive eigenvalues $8h^2V^{-1}(1+o(1))$, and a uniform complementary gap. These estimates hold for every simultaneous path $p = p(h) \in \mathcal{B}^\circ$ tending to the conifold, without a comparison between the base and bundle rates.

Proof. Diagonal and off-diagonal entries are orthogonal for both forms. For diagonal entries $x_j = S_{jj}$, $y_j = T_{jj}$, set $a_j = (x_j - y_j)/\sqrt{2}$ and $b_j = (x_j + y_j)/\sqrt{2}$. The trace condition is $b_1 + b_2 = 0$, and

$$E_{\mathrm{diag}} = 8d^2a_1^2 + 8e^2a_2^2, \qquad M_{\mathrm{diag}} = V\sum_j (a_j^2 + b_j^2).$$

For off-diagonal entries $u = S_{12}$ and $v = T_{12}$,

$$E_{\mathrm{off}} = 4(d^2+e^2)(|u|^2+|v|^2) - 16de\,\mathrm{Re}(\bar{u}v),$$
$$M_{\mathrm{off}} = 2V(|u|^2+|v|^2).$$

The real two-dimensional spaces $u = v$ and $u = -v$ give the two double eigenvalues in (90). This proves the complete list, including all special zero multiplicities. Along the stated path $|z| = O(h)$ uniformly, because $c_\pm$ stay positive and bounded. Proposition 7.1 therefore transfers the list with relative error $O(h)$, independently of the ratio h^{2b}/h^2. □

Figure 1 also illustrates why the size of the arrows alone does not determine the weakest rates. The linear relaxation time of the two difference modes is of order h^{-2b}. An approximation of nonlinear trajectories on such intervals requires more than the spectral comparison.

The next three sections construct the exact finite evolution and the phase selected by a spatially varying initial metric.

9. Heat smoothing and entry from spatial initial metrics

Fix a smooth base point p and an actual HE representative w from Proposition 7.1. Throughout the flow argument these parameters remain fixed. Write $r = |w|$, $D = D_{\mathcal{A}_p(w)} = D_0 + a$ and $L = D^*D$, $L_0 = D_0^*D_0$. All section variables are Hermitian and trace free for the fixed transported metric h_0. Let Π be the split projection onto $\mathfrak{h}$ and $Q = I - \Pi$. On each smooth fibre put

$$Y_p = C^0(\mathrm{Herm}_0\, E^{\mathrm{sp}}), \qquad X_p = C^1(\mathrm{Herm}_0\, E^{\mathrm{sp}}), \qquad \|u\|_X = \|u\|_\infty + \|D_0 u\|_\infty. \tag{91}$$

We use these fibrewise spaces with uniform operator bounds; no fixed Banach bundle through the singular fibre is required.

Theorem 9.1. *There are $M, \sigma, C, \delta, r_0 > 0$, independent of the base scales and of all positive weak-rate ratios, such that*

$$\begin{aligned} \|e^{-tL_0}Q\|_{X_p \to X_p} &\le M e^{-\sigma t}, \\ \|e^{-tL_0}Q\|_{Y_p \to X_p} &\le M(1 + t^{-1/2})e^{-\sigma t} \quad (t > 0). \end{aligned} \tag{92}$$

In the metric coordinate $h = h_0 e^u$, the actual HYM equation is

$$\dot{u} + L_0 u = N_w(u). \tag{93}$$

Here $N_w : X_p \to Y_p$ depends only on the first spatial jet, and is smooth with uniform derivatives of every fixed finite order on a common small ball. For parallel S and $v \in QX_p$ with $|S|, \|v\|_X \le \delta_1$, it satisfies

$$\begin{aligned} \|N_w(S)\|_\infty &\le C r^2 |S|, \\ \|N_w(S+v) - N_w(S)\|_\infty &\le C(r + \delta_1)\|v\|_X. \end{aligned} \tag{94}$$

Every fixed positive-order derivative of $N_w(S)$ in S is $O(r^2)$ on this ball.

For every smooth u_0 with $\|u_0\|_X < \delta$ and $r < r_0$, the determinant-normalized actual flow exists for all $t \ge 0$, stays in a common small C^1 ball, and obeys

$$\begin{aligned} \|u(t)\|_\infty &\le \|u_0\|_\infty, \qquad |\Pi u(t)| \le \|u_0\|_\infty, \\ \|Qu(t)\|_X &\le C e^{-\sigma t/2}\|Qu_0\|_X + C r^2 \|u_0\|_\infty. \end{aligned} \tag{95}$$

For $0 < r < \min(r_0, 1)$, the time $T_r = 4\sigma^{-1}\log(1/r)$ gives $\|Qu(t)\|_X \le C r^2 \delta$ for all $t \ge T_r$. Every positive weak eigenvalue satisfies $T_r \lambda_j \to 0$ uniformly as $r \to 0$.

9.1. Uniform first derivatives of the split heat flow.

Let $S_t = e^{-tL_0}$ and let H_t denote scalar heat flow. Kato domination gives $|S_t f| \le H_t |f|$, so $\|S_t f\|_\infty \le \|f\|_\infty$. The scalar estimates from Section 5 and Bochner's formula with $\mathrm{Ric}(g_p) = 0$ give

$$\|\mathrm{d}H_t\phi\|_\infty \le C t^{-1/2}\|\phi\|_\infty, \qquad \|\mathrm{d}H_t\phi\|_\infty \le C\|\mathrm{d}\phi\|_\infty. \tag{96}$$

Use the fixed finite cover and unitary line frames from the proof of Lemma 5.1. Their connection coefficients are bounded and divergence free. The partition functions χ_j have uniformly bounded gradients and Laplacians; the derivatives of the core cutoffs are supported in fixed exterior annuli. If $f_j(t)$ is a scalar coefficient of $S_t f$ in one frame, extend $z_j = \chi_j f_j$ by zero to the whole fibre. The local formula (43) gives

$$(\partial_t + \Delta) z_j = F_j, \qquad \|F_j(t)\|_\infty \le C\{\|D_0 S_t f\|_\infty + \|f\|_\infty\}.$$

Conversely, $\|D_0 S_t f\|_\infty \le C\sum_j \|\mathrm{d}z_j\|_\infty + C\|f\|_\infty$. No derivative of the full curvature enters these formulas.

Scalar Duhamel and (96) therefore give, for $G(t) = \|D_0 S_t f\|_\infty$ and $F = \|f\|_\infty$,

$$G(t) \le C(t^{-1/2}+1)F + C\int_0^t (t-s)^{-1/2}\{G(s)+F\}\,\mathrm{d}s \quad (0<t\le 1).$$

For smooth f, set $Z_T = \sup_{0<t\le T}\sqrt{t}\,G(t)$. The identity $\int_0^t (t-s)^{-1/2}s^{-1/2}\,\mathrm{d}s = \pi$ implies $Z_T \le CF + C\sqrt{T}Z_T$. One common small T absorbs the last term. The semigroup law extends the bound to fixed bounded time intervals. For $f \in C^1$, use the second scalar estimate on the initial term instead. Taking $\sup_{0<t\le T} G(t)$ and using $\int_0^t (t-s)^{-1/2}\,\mathrm{d}s = 2\sqrt{t}$ yields

$$\|S_t f\|_X \le C\|f\|_X \quad (0\le t\le 1), \qquad \|S_t f\|_X \le C(1+t^{-1/2})\|f\|_\infty \quad (0<t\le 1).$$

Density extends these estimates. On each fixed smooth fibre, the smooth heat solutions converge to their initial data in C^1; density and the first estimate prove strong continuity on X_p. The same argument works on Y_p.

The averaging projection Π is uniformly bounded on both spaces and commutes with S_t. The split L^2 gap, fixed volume, and the fixed-time L^2-to-L^∞ bound used in Lemma 5.1 give $\|S_t Q\|_{Y_p\to Y_p} \le Ce^{-ct}$ for $t \ge 1$. Composing with a fixed positive-time Y_p-to-X_p map proves (92). This argument supplies the first-derivative smoothing needed below without a uniform higher-order elliptic estimate.

9.2. **The contracted connection derivative.** The connection expansion already gives $\|a\|_\infty \le Cr$. We also need its divergence:

$$\|D_0^* a\|_\infty \le Cr^2. \tag{97}$$

Indeed put $U = e^{s_p(w)/2}$ and $\gamma = Ub_p(w)U^{-1} - (\bar\partial_0 U)U^{-1}$, so $a = \gamma - \gamma^*$. Section 5 gives

$$\|b_p(w)\|_\infty \le Cr, \quad \|\bar\partial_0^* b_p(w)\|_\infty \le Cr^2, \quad \|D_0 U\|_\infty + \|P_p U\|_\infty \le Cr^2.$$

The same first-derivative estimate holds for U^{-1}. The divergence product rule gives

$$\bar\partial_0^*(UbU^{-1}) = U(\bar\partial_0^* b)U^{-1} + O_\infty(|D_0U||b| + |b||D_0U^{-1}|),$$
$$\bar\partial_0^*((\bar\partial_0 U)U^{-1}) = (P_pU)U^{-1} + O_\infty(|D_0U||D_0U^{-1}|).$$

Thus $\|\bar\partial_0^*\gamma\|_\infty \le Cr^2$. Real unitary divergence commutes with adjoints, proving (97). For $A_j = \mathrm{ad}_{a(e_j)}$ in a real orthonormal frame, expansion of the connection Laplacian gives

$$(L - L_0)u = -2\sum_j A_j D_{0,e_j} u + [D_0^* a, u] - \sum_j A_j^2 u. \tag{98}$$

Its first-order coefficient is $O(r)$ and its zeroth-order coefficient is $O(r^2)$.

9.3. **The exact equation in logarithmic coordinates.** Define the pointwise invertible linear map

$$J_u(V) = \int_0^1 e^{-\theta u} V e^{\theta u}\,\mathrm{d}\theta.$$

It satisfies $e^{-u}(\mathrm{d}\exp)_u(V) = J_u(V)$. Diagonalization of u shows that its scalar multipliers are $(1-e^{-a})/a$, with value one at $a = 0$. Its inverse and all fixed-order derivatives are bounded on a fixed small pointwise ball. Since the reference connection D is HE, the exact Chern formula is

$$\mathrm{i}\Lambda F_{h_0 e^u} = \mathrm{i}\Lambda\bar\partial_D(J_u(D'_D u)) = J_u(\mathrm{i}\Lambda\bar\partial_D D'_D u) + \mathrm{i}\Lambda(dJ)_u[\bar\partial_D u]\wedge D'_D u.$$

The section Kähler identity gives $\mathrm{i}\Lambda\bar\partial_D D'_D u = Lu/2$. Consequently $h^{-1}\dot h = -2\mathrm{i}\Lambda F_h$ becomes

$$\begin{aligned} \dot u + Lu &= \mathcal{Q}(u, Du), \\ \mathcal{Q}(u, Du) &= -2J_u^{-1}\mathrm{i}\Lambda(dJ)_u[\bar\partial_D u]\wedge D'_D u. \end{aligned} \tag{99}$$

Here dJ differentiates its finite matrix coefficient u. The nonlinear term is quadratic in the first spatial jet; all second derivatives have cancelled. Thus

$$N_w(u) = -(L - L_0)u + \mathcal{Q}(u, D_0 u + [a, u]) \tag{100}$$

has the asserted uniform smoothness from X_p to Y_p. This follows from bounded matrix derivatives and the continuous pointwise product $C^0 \times C^0 \to C^0$. The equation preserves Hermitian variables by its metric origin. It preserves trace zero because the determinant metric solves scalar heat and initially equals the fixed HE determinant.

For parallel S, the first-order term in (98) vanishes, and $DS = [a, S]$. Hence $N_w(S) = O(r^2|S|) + O(r^2|S|^2)$, with the same factor r^2 in every fixed finite derivative. Taking the difference at $S + v$ proves (94).

9.4. **Confinement and normal decay.** On each fixed compact smooth fibre, global smooth HYM metric flow from a smooth positive metric follows from [Kob87, Theorem 6.8.25]. We use this only for existence; its higher constants need not be uniform. The squared pointwise distance from h_0 to $h_0 e^u$ in the space of positive Hermitian metrics is $\operatorname{tr} u^2$. The two-solution distance inequality [Kob87, Lemma 6.4.5], with time changed by $\tau = 2t$ to our convention, gives

$$(\partial_t + \Delta)\tfrac{1}{2}\operatorname{tr} u^2 \le 0. \tag{101}$$

The reference HE metric is stationary. The maximum principle therefore proves the first bound in (95). Block averaging and Jensen's inequality give the bound for $S(t) = \Pi u(t)$.

Let $v = Qu$ and $\varepsilon = \|u_0\|_\infty$. As long as $\|v\|_X \le \delta_1$, Duhamel gives

$$\|v(t)\|_X \le Me^{-\sigma t}\|v(0)\|_X + \int_0^t k(t-s)\{a_0\|v(s)\|_X + b_0\}\,\mathrm{d}s, \tag{102}$$

where $k(t) = M'(1 + t^{-1/2})e^{-\sigma t}$, $a_0 = C(r + \delta_1)$, and $b_0 = Cr^2\varepsilon$. Choose δ_1, r_0 so that $a_0\int_0^\infty k(t)e^{\sigma t/2}\,\mathrm{d}t \le 1/2$. Iteration of this nonnegative convolution inequality gives

$$\|v(t)\|_X \le 2Me^{-\sigma t/2}\|v(0)\|_X + 2b_0\int_0^\infty k(s)\,\mathrm{d}s.$$

Equivalently the right side is a supersolution: its exponential term is controlled by the weighted kernel integral and its constant term by the unweighted integral. Uniform boundedness of Π, Q lets us choose one initial δ making this bound smaller than $\delta_1/2$. A first-exit argument then proves it for all time. Since $D_0 S = 0$, this also confines u in the required C^1 ball and completes (95).

At T_r the exponential is at most r^2. Theorem 6.1 gives $\lambda_j \le Cr^2$ for every positive weak rate, so $T_r\lambda_j \le Cr^2\log(1/r) \to 0$. This proves Theorem 9.1, including arbitrarily unequal weak rates. At $r = 0$ there are no positive weak rates, and the normal part decays exponentially.

The basin is an open ball in the C^1 topology intersected with smooth determinant-normalized metrics. It contains spatially varying initial metrics. The persistent $O(r^2)$ normal term in (95) does not by itself give tracking on a long time interval. We next absorb that term into an invariant finite-dimensional graph.

10. An invariant graph and its exact finite gradient flow

The stationary metric correction of Section 5 does not define an invariant graph for metric evolution. We construct such a graph directly from the heat estimates of Section 9. Fix a finite regularity order $m \geq 4$. The constants and the chosen graph may depend on m.

Theorem 10.1. *After a common shrink there is a chosen C^m map $\psi_{p,w} : \mathfrak{h} \to QX_p$, with $\psi_{p,w}(0) = 0$, such that*

$$\sup_S \|D_S^j \psi_{p,w}(S)\|_X \leq C_m r^2 \quad (0 \leq j \leq m). \tag{103}$$

For $j \leq m-1$ the same bound holds for $L_0 D_S^j \psi$ in Y_p. The graph is invariant for the extension defined below and locally invariant for actual HYM. Its complete finite field $f_{p,w}$ satisfies $\|f_{p,w}\|_{C^m} \leq C_m r^2$. On a fixed inner ball, a trajectory of this field reconstructs the actual metric flow

$$h(t) = h_0 \exp\{S(t) + \psi_{p,w}(S(t))\}. \tag{104}$$

The graph points there are spatially smooth on each fixed fibre. At zero the tangent map $J_0 = I + D\psi_{p,w}(0)$ identifies $\mathfrak{h}$ with the complete seven-dimensional small spectral space of L, and

$$J_0 D f_{p,w}(0) = -L J_0. \tag{105}$$

All exact kernel directions are retained. The construction is jointly K-equivariant in w and S.

10.1. **A smooth extension of the first-jet equation.** Choose smooth scalar cutoffs θ, χ taking values in $[0,1]$, equal to one on $[0,1]$ and zero on $[4,\infty)$. For a fixed small $a > 0$, put

$$\widehat{S} = \chi(|S|^2/a^2)S, \qquad \vartheta(v)(x) = \theta\big((|v(x)|^2 + |D_0 v(x)|^2)/a^2\big).$$

The extended nonlinearity is

$$\widetilde{N}_w(S,v)(x) = \vartheta(v)(x) N_w(\widehat{S} + v)(x). \tag{106}$$

Its right side is evaluated using the pointwise formula (100), with $D_0 \widehat{S} = 0$. Choose a so that the support of the cutoff lies strictly inside that formula's pointwise domain, and extend by zero outside. This uses a smooth cutoff in finite-dimensional first-jet variables. It requires no smooth bump function of the C^1 norm.

The bounded jet map $v \mapsto (v, D_0 v)$ and the pointwise chain rule make $\widetilde{N}_w : \mathfrak{h} \times QX_p \to Y_p$ globally smooth and bounded, with uniform fixed-order derivative bounds. On the cutoff support the drift term has size $O(ra)$ and the quadratic term has size $O(a^2 + r^2 a^2)$. A cutoff derivative contributes at most C/a. For $r \leq a$ this proves

$$\|D\widetilde{N}_w\| \leq \ell = C(a + r), \qquad \sup_S \|D_S^j \widetilde{N}_w(S,0)\|_Y \leq C_j r^2. \tag{107}$$

The second bound follows because every occurrence of a parallel variable in (100) retains two connection factors or the zeroth-order coefficient. Also $\widetilde{N}_w(0,0) = 0$. Set $\mathcal{A}(S,v) = \Pi \widetilde{N}_w(S,v)$ and $\mathcal{B}(S,v) = Q\widetilde{N}_w(S,v)$, absorbing the uniform projection norms into ℓ. The extended system is

$$\dot{S} = \mathcal{A}(S,v), \qquad \dot{v} = -L_0 v + \mathcal{B}(S,v). \tag{108}$$

It agrees with actual HYM when $|S| < a$ and $\|v\|_X < a$. The integrable smoothing kernel and global Lipschitz bounds give unique global forward mild solutions on each fibre. All cutoffs use invariant norms, so the extension is jointly equivariant under parallel unitary conjugation.

10.2. **Backward histories and finite differentiability.** Let $k(t) = M(1+t^{-1/2})e^{-\sigma t}$ and $K_\beta = \int_0^\infty k(t)e^{\beta t}\,\mathrm{d}t$ for $0 \le \beta < \sigma$. For continuous histories on $(-\infty, 0]$ use the norm

$$\|(S,v)\|_\beta = \sup_{t\le 0} e^{\beta t}(|S(t)| + \|v(t)\|_X).$$

Put $\eta = \sigma/(2(m+2))$. Reduce a and then r_0 so that

$$\ell(\beta^{-1} + K_\beta) < \tfrac{1}{2} \quad \big(\eta \le \beta \le (m+1)\eta\big), \qquad \ell K_0 < \tfrac{1}{2}. \tag{109}$$

For any endpoint S_0, solve

$$\begin{aligned} S(t) &= S_0 - \int_t^0 \mathcal{A}(S(s), v(s))\,\mathrm{d}s, \\ v(t) &= \int_{-\infty}^t e^{-(t-s)L_0}\mathcal{B}(S(s), v(s))\,\mathrm{d}s. \end{aligned} \tag{110}$$

The two Lipschitz bounds are ℓ/β and ℓK_β. Boundedness of the nonlinearities shows that the map sends each indicated history space into itself: the finite output grows at most linearly backwards, and the normal output is bounded. Continuity of the latter in X_p follows by splitting off a short terminal interval and using integrability of k. Thus there is a unique fixed history, identical in all these weighted spaces. Its normal part satisfies

$$\sup_{t\le 0}\|v(t)\|_X \le K_0\{Cr^2 + \ell \sup_{t\le 0}\|v(t)\|_X\} \le Cr^2.$$

Define $\psi(S_0) = v(0)$. The zero history gives $\psi(0) = 0$.

We record the differentiability argument to account for products of growing histories. The first variation solves the linearized fixed-point equation with initial term $(\delta S_0, 0)$. Its integral operator has norm below $1/2$ in each weight. Differences of fixed histories are $O(|\delta S_0|e^{-\eta t})$, so the quadratic Taylor remainder is $O(|\delta S_0|^2 e^{-2\eta t})$. Inverting in the 2η space proves actual differentiability. At order j, the inhomogeneous terms have the form

$$D^q(\mathcal{A},\mathcal{B})(S,v)[Z_{i_1},\dots,Z_{i_q}], \qquad q \ge 2, \quad i_1 + \cdots + i_q = j,$$

with each $i_k < j$. Their weights add to $j\eta$. The same inverse works there, and Taylor's formula with the next derivative in the spare weight $(j+1)\eta$ proves differentiability and continuity. All these weights are below σ. Evaluation at zero proves C^m regularity of ψ with uniform derivative bounds.

For the sharper r^2 bound, bounded mixed derivatives, the normal bound, and (107) give $D_S^j(\mathcal{A},\mathcal{B}) = O(r^2)$ along every fixed history. In the first normal variation, the finite coefficient is therefore $O(r^2)$ and the normal coefficient is small; absorption makes that variation $O(r^2)$. At higher orders, a term with only finite variations has an $O(r^2)$ coefficient, while every other term contains an $O(r^2)$ normal variation. Induction in the weighted equations proves (103). Uniqueness also proves joint equivariance.

10.3. **Invariance, domains and linearization.** Shift a fixed history by a finite time, attaching the forward mild solution when necessary. Its negative tail still has bounded normal component and at most linear finite growth. The shifted normal integral has no term at minus infinity. It therefore solves (110) with its new endpoint. Uniqueness gives $v(t) = \psi(S(t))$. Thus the graph is invariant and its field is

$$f(S) = \mathcal{A}(S, \psi(S)). \tag{111}$$

The chain rule and the preceding estimates give $\|f\|_{C^m} \leq C_m r^2$. This bounded field has a complete finite flow in both time directions.

On the graph the mild equation and differentiation in X_p imply the generator-domain identity

$$L_0\psi(S) = \mathcal{B}(S, \psi(S)) - D\psi(S)f(S). \tag{112}$$

Indeed divide the mild formula at time t minus its initial value by t and let $t \downarrow 0$ in Y_p. Strong continuity makes the integral quotient converge to $\mathcal{B}$ and the left derivative equal $D\psi f$. This is the definition of the generator domain. Differentiating the right side and using closedness of L_0 gives the claimed domain bounds through order $m - 1$.

On a fixed smooth fibre, ordinary elliptic regularity now gives $\psi, D\psi \in W^{2,q}$ for every finite q, then $C^{1,\alpha}$. Equation (112) gives $C^{2,\alpha}$ regularity. In an inner finite ball, a short graph trajectory stays where the cutoffs are inactive, hence is a classical solution of the original first-jet equation. Interior fixed-fibre parabolic regularity makes it spatially smooth. These higher regularity statements have no asserted uniform constants. They justify the actual reconstruction (104).

At the HE origin $DN_w(0) = -(L - L_0)$. Differentiating invariance gives (105). The seven-dimensional image of J_0 is invariant under the self-adjoint L and lies in its operator domain. Its restriction is self-adjoint for the induced mass; the bound on $Df(0)$ puts all its eigenvalues in $[0, Cr^2]$. The complementary gap in Theorem 6.1 then identifies this image with the whole small spectral space. Since $\Pi J_0 = I$, its coordinates are exactly those of the inverse restricted projection.

If S is in the constant actual parallel kernel, then $DS = 0$ and $h_0 e^S$ is HE. Its constant history gives $\psi(S) = f(S) = 0$ in the inner ball. Conversely a finite equilibrium reconstructs an HE metric. Integrating the squared-distance identity in the proof of [Kob87, Lemma 6.4.5] along the fibrewise geodesic between two stationary HE metrics forces their logarithmic difference to be D-parallel. Theorem 6.1 identifies that difference with a constant element of $\mathfrak{h}$. Thus the local equilibrium set is exactly the actual parallel kernel ball. This completes the proof of Theorem 10.1.

10.4. **The finite metric, energy and full Hessian.** Put $Z(S) = S + \psi(S)$ and $Z_S a = a + D\psi(S)a$. The pullback of the invariant L^2 metric on positive Hermitian metrics is

$$g_{\mathrm{fin},S}(a, b) = \int_{X_p} \mathrm{tr}\{J_{Z(S)}(Z_S a) J_{Z(S)}(Z_S b)\}\, \mathrm{d}V_{g_p}. \tag{113}$$

Although the factors need not be h_0-Hermitian, they are Hermitian for the current metric, so this is positive. On the small ball it is uniformly equivalent to the L^2 trace pairing of $Z_S a, Z_S b$, hence to M since $\Pi Z_S a = a$. Define

$$\mathcal{E}_{\mathrm{fin}}(S) = 2 \int_0^1 \int_{X_p} \mathrm{tr}\{\mathrm{i}\Lambda F_{h_0 e^{tZ(S)}} Z(S)\}\, \mathrm{d}V_{g_p}\, \mathrm{d}t. \tag{114}$$

This is twice Donaldson's functional relative to h_0. Its first variation, with $\mathrm{d}V = \omega^3/3!$, is [Kob87, Section 6.3, equations (6.3.25)–(6.3.27)]

$$\mathrm{d}\mathcal{D}_h(\dot{h}) = \int_{X_p} \mathrm{tr}(\mathrm{i}\Lambda F_h\, h^{-1}\dot{h})\,\mathrm{d}V.$$

On the inner region where the graph carries the actual HYM equation, graph invariance and its coefficient -2 give

$$f = -\operatorname{grad}_{g_{\mathrm{fin}}} \mathcal{E}_{\mathrm{fin}}. \tag{115}$$

This metric describes variations of metrics on one fixed holomorphic bundle. The harmonic metric on varying bundle moduli will be constructed separately.

Corollary 10.2. *At an HE parameter $w = \chi_p(z)$, the exact finite mass and Hessian satisfy*

$$\begin{aligned} g_{\mathrm{fin},0}(S,T) &= M(S,T) + \langle D\psi(0)S, D\psi(0)T\rangle_2, \\ |g_{\mathrm{fin},0} - M| &\le C|z|^4 M, \\ (1-C|z|)E_{p,z} &\le \mathrm{Hess}_0\,\mathcal{E}_{\mathrm{fin}} \le (1+C|z|)E_{p,z}. \end{aligned} \tag{116}$$

In particular the Hessian has the exact kernel of $E_{p,z}$, and its bilinear error is bounded by $C|z|\sqrt{E_{p,z}(S)E_{p,z}(T)}$.

Proof. At the critical origin the exact forms are $N(S,T) = \langle J_0 S, J_0 T\rangle_2$ and $H(S,T) = \langle LJ_0S, J_0T\rangle_2$. Orthogonality gives the mass formula and its $O(r^4)$ bound. The lower bound in (57) gives $H \ge (1-Cr)E_{p,w}$. Their kernels agree because J_0 is the inverse of Π on the full small spectral space and the actual kernel consists of the constant stabilizer directions. On the quotient by that common kernel, let $\bar{N}, \bar{M}$ be the quotient masses (the infima over representatives). They are relatively $1+O(r^4)$ equivalent. The positive generalized eigenvalues of (H,N) are exactly the positive small eigenvalues of L. Therefore (59) gives

$$\frac{\det \bar{H}}{\det \bar{E}_{p,w}} = \frac{\prod_{\lambda_j>0}\lambda_j}{\prod_{\mu_j>0}\mu_j}\frac{\det \bar{N}}{\det \bar{M}} = 1 + O(r).$$

Every eigenvalue of $\bar{E}_{p,w}^{-1/2}\bar{H}\bar{E}_{p,w}^{-1/2}$ is at least $1-Cr$, and their product is $1+O(r)$. Since the dimension is at most seven, each is at most $1+Cr$. If the quotient has dimension zero the assertion is immediate. This proves the full relative form comparison without a lower bound for a positive weak rate. Finally equivariance of χ_p gives $\delta_w S = d\chi_p(z)\delta_z S$, with derivative $I + O(|z|)$ in the uniformly positive arrow norm. Hence $E_{p,w} = (1+O(|z|))E_{p,z}$ as forms, including kernels. Polarization on the quotient proves the bilinear estimate. □

The exact linear generator is $-g_{\mathrm{fin},0}^{-1}H$. The relative Hessian statement does not replace the nonlinear field (111) by its quadratic approximation on arbitrarily long time intervals.

11. The spatial phase and all-time metric comparison

Fix one graph from Theorem 10.1, and write ϕ_t for its complete finite flow. The phase of a general initial metric will be defined by removing this finite evolution from its parallel component.

Theorem 11.1. *There are common $r_0, \delta, c, C > 0$ and a C^1 map $\Gamma_{p,w} : B_{X_p}(\delta) \to \mathfrak{h}$, with uniformly bounded first derivative, having the following properties. For every smooth initial metric in this ball, let $u(t)$ be the actual flow and put*

$$S(t) = \Pi u(t), \qquad y(t) = Qu(t) - \psi(S(t)), \qquad D_* = \|y(0)\|_X.$$

Then

$$\begin{aligned}\Gamma(u_0) &= \lim_{T\to\infty} \phi_{-T}(S(T)),\\ S_{\mathrm{eff}}(t) &= \phi_t(\Gamma(u_0)), \qquad u_{\mathrm{eff}}(t) = S_{\mathrm{eff}}(t) + \psi(S_{\mathrm{eff}}(t))\end{aligned} \tag{117}$$

are defined for all $t \geq 0$. The reconstructed metric $h_{\mathrm{eff}} = h_0 e^{u_{\mathrm{eff}}}$ is an actual HYM solution on the same fixed holomorphic bundle, and

$$\|y(t)\|_X + \|u(t) - u_{\mathrm{eff}}(t)\|_X \leq C D_* e^{-ct}. \tag{118}$$

The centered relative logarithm

$$\zeta(t) = \log\{e^{-u_{\mathrm{eff}}(t)/2} e^{u(t)} e^{-u_{\mathrm{eff}}(t)/2}\} \tag{119}$$

satisfies $\|\zeta(t)\|_X \leq C D_* e^{-ct}$, and hence

$$e^{-CD_* e^{-ct}} h_{\mathrm{eff}}(t) \leq h(t) \leq e^{CD_* e^{-ct}} h_{\mathrm{eff}}(t). \tag{120}$$

All constants are independent of the base scales, stabilizer type, and ratios of positive weak rates. One can take $c = \sigma/2$, so the error is at most $Cr^2 D_*$ after $T_r = 4\sigma^{-1}\log(1/r)$.

The phase equals S_0 on graph data $S_0 + \psi(S_0)$. It is unique among complete finite trajectories that track $S(t)$ at an exponential rate greater than $L_f = \|Df\|_\infty \leq Cr^2$. It is jointly K-equivariant and depends on the chosen graph and the full initial metric.

11.1. **Normal decay and the phase limit.** In the coordinates S, y, the extended equation is exactly

$$\dot{S} = f(S) + h(S,y), \qquad \dot{y} = -L_0 y + F(S,y), \tag{121}$$

where

$$\begin{aligned}h(S,y) &= \mathcal{A}(S, \psi(S)+y) - \mathcal{A}(S,\psi(S)),\\ F(S,y) &= \mathcal{B}(S,\psi(S)+y) - \mathcal{B}(S,\psi(S)) - D\psi(S)h(S,y).\end{aligned}$$

The final term follows from differentiating the graph coordinate and is essential. These maps vanish at $y = 0$. Their uniform derivatives satisfy

$$\|D_y h\| + \|D_y F\| \leq \varepsilon, \qquad \|D_S h(S,y)\| + \|D_S F(S,y)\| \leq C\|y\|_X, \tag{122}$$

where ε is made small by choosing the cutoff radius and then r_0. The second bound follows from bounded mixed derivatives and vanishing at $y = 0$. For either sign of t, the finite variational equations give

$$\|D\phi_t\| \leq e^{L_f|t|}, \qquad \|D^2\phi_t\| \leq K_f|t| e^{2L_f|t|}, \qquad K_f = \|D^2 f\|_\infty = O(r^2). \tag{123}$$

Set $c = \sigma/2$. Choose the constants so that $\varepsilon \int_0^\infty k(t) e^{ct}\,\mathrm{d}t < 1/2$. The normal Duhamel formula then gives

$$\sup_{s\leq T} e^{cs}\|y(s)\|_X \leq M D_* + \tfrac{1}{2}\sup_{s\leq T} e^{cs}\|y(s)\|_X.$$

It follows that $\|y(t)\|_X \leq 2MD_* e^{-ct}$. This prevents normal exit for small initial D_*; the bounded finite field and the integrable perturbation h prevent finite-time escape of S.

For $P_T = \phi_{-T}(S(T))$, differentiation cancels the finite f terms and gives

$$\frac{\mathrm{d}P_T}{\mathrm{d}T} = D\phi_{-T}(S(T))h(S(T), y(T)). \tag{124}$$

Its norm is at most $C\varepsilon D_* e^{-(c-L_f)T}$. Taking $L_f < c$ proves the phase limit and $|\Gamma - S(0)| \le C\varepsilon D_*$. For fixed t, apply the same differentiation to $\phi_{t-T}(S(T))$ for $T \ge t$. Integration to infinity gives

$$|S(t) - S_{\mathrm{eff}}(t)| \le C\varepsilon D_* e^{-ct}. \tag{125}$$

Together with the normal decay and $D\psi = O(r^2)$ this proves (118) for the extended system. If two finite trajectories track at rates larger than L_f, applying ϕ_{-t} to their difference forces their initial points to agree. Graph data have $y = 0$, so their phase is their initial finite coordinate. This argument uses an upper expansion bound L_f; it never divides by a positive weak rate.

11.2. **Differentiability of the phase.** The integrable smoothing kernel gives differentiable dependence of mild solutions on each finite interval. Let $W = \delta S$ and $Z = \delta y$. Their equations are

$$\begin{aligned} \dot W &= (Df + D_S h)W + D_y h\, Z, \\ Z(t) &= e^{-tL_0} Z(0) + \int_0^t e^{-(t-s)L_0}(D_S F\, W + D_y F\, Z)(s)\,\mathrm{d}s. \end{aligned}$$

Put $\gamma = \sigma/16$ and reduce r_0 so that $L_f < \gamma/4$. With $U_T = \sup_{t\le T} e^{-\gamma t}|W(t)|$ and $V_T = \sup_{t\le T} e^{(c-\gamma)t}\|Z(t)\|_X$, normal convolution and (122) give

$$V_T \le C\|Z(0)\|_X + CD_* U_T, \qquad U_T \le C|W(0)| + C\varepsilon V_T.$$

In the second inequality the coefficient $D_S h$ is integrable in time. For sufficiently small D_*, absorption gives

$$\begin{aligned} |W(t)| &\le Ce^{\gamma t}(|W(0)| + \varepsilon\|Z(0)\|_X), \\ \|Z(t)\|_X &\le Ce^{-(c-\gamma)t}(\|Z(0)\|_X + D_*|W(0)|). \end{aligned} \tag{126}$$

The derivative of the velocity (124) is

$$D^2\phi_{-T}(S(T))[W(T), h] + D\phi_{-T}(S(T))(D_S h\, W(T) + D_y h\, Z(T)).$$

Its operator norm is bounded by a constant times

$$\varepsilon D_*(1+T)e^{-(c-2L_f-\gamma)T} + (D_* + \varepsilon)e^{-(c-L_f-\gamma)T}.$$

This is uniformly integrable on a smaller initial ball. The integral formula for P_T therefore converges in C^1, with continuous derivative by dominated convergence. Composing with $S(0) = \Pi u_0$ and $y(0) = Qu_0 - \psi(\Pi u_0)$ proves the asserted phase regularity. Equivariance follows from that of the equations and uniqueness of the phase.

11.3. **Confinement in the actual equation.** Theorem 9.1 lets us reduce δ so that every smooth actual solution in the initial ball satisfies $|S(t)| < a/4$ and $\|Qu(t)\|_X < a/4$ for all positive time. Thus it remains where the extended and actual equations agree. Local mild uniqueness identifies the two solutions globally. Also $D_* \le \|Qu_0\|_X + Cr^2|\Pi u_0| \le C\delta$. Reduce δ again so that (125) is less than $a/4$. Then $|S_{\mathrm{eff}}(t)| < a/2$, and the graph bound gives $\|\psi(S_{\mathrm{eff}}(t))\|_X < a/4$. The selected graph trajectory also stays in the actual region for all time. Theorem 10.1 reconstructs its spatially smooth actual HYM metric there.

The norm X_p is a Banach algebra norm, since D_0 is a derivation. Exponential and logarithm on a fixed small positive matrix ball have bounded derivatives. Hence

$$(u, \widetilde u) \longmapsto \log(e^{-\widetilde u/2} e^{u} e^{-\widetilde u/2})$$

is uniformly locally Lipschitz from $X_p \times X_p$ to X_p and vanishes on the diagonal. This proves the bound for (119). Its matrix is positive Hermitian with determinant one, so its spectral bounds imply (120). These metric inequalities transport invariantly back to the actual holomorphic bundle. The stated C^1 norm is the one in the fixed transport used throughout the construction. At T_r we have $e^{-cT_r} = r^2$. This completes the proof of Theorem 11.1.

11.4. **Finite preparation and all weak time scales.** The limit defining the phase has a finite preparation rule. For any $T \geq 0$, put $\Gamma_T = \phi_{-T}(\Pi u(T))$ and let $u_{\mathrm{eff},T}$ be the graph trajectory starting at Γ_T. Integrating (124) gives

$$|\Gamma_T - \Gamma| \leq C\varepsilon D_* e^{-(c-L_f)T}, \qquad |\Gamma_T - S(0)| \leq C\varepsilon D_*. \tag{127}$$

All these initial points can be put in a common graph ball whose positive trajectories are actual and confined: for small S_1, $\|S_1 + \psi(S_1)\|_X \leq C|S_1|$, so Theorem 9.1 applies, and uniqueness preserves the graph. A smaller initial δ puts Γ and every Γ_T in that ball.

The two-solution distance inequality [Kob87, Lemma 6.4.5], applied to these two actual graph flows, implies

$$\sup_x d_x(h_{\mathrm{eff},T}(t), h_{\mathrm{eff}}(t)) \leq \sup_x d_x(h_{\mathrm{eff},T}(0), h_{\mathrm{eff}}(0)) \leq C|\Gamma_T - \Gamma|.$$

Here d_x is the invariant distance on positive Hermitian metrics in the fibre. On the common small ball it is equivalent to the pointwise difference in logarithmic coordinates. Apply the bounded projection Π to obtain $|S_T(t) - S_{\mathrm{eff}}(t)| \leq C|\Gamma_T - \Gamma|$. The graph derivative bound upgrades this finite-coordinate bound to

$$\sup_{t\geq 0} \|u_{\mathrm{eff},T}(t) - u_{\mathrm{eff}}(t)\|_X \leq C\varepsilon D_* e^{-(c-L_f)T}. \tag{128}$$

The upgrade uses the finite graph; it does not assert C^1 contraction for arbitrary metric flows. Combining with the actual tracking estimate gives

$$\left\| \log\{ e^{-u_{\mathrm{eff},T}(t)/2} e^{u(t)} e^{-u_{\mathrm{eff},T}(t)/2} \} \right\|_X \leq CD_* e^{-ct} + C\varepsilon D_* e^{-(c-L_f)T}.$$

Since $L_f = O(r^2)$, taking $T = T_r$ makes the second term $O(r^2 D_*)$ uniformly for every later comparison time. Moreover $T_r \lambda_j \to 0$ for every positive weak rate. Thus the preparation is negligible compared with all inverse weak rates, including the times h^{-2b} in Proposition 8.2.

The comparison is absolute tracking to the exact corrected field f. It does not supply a relative error divided by a vanishing trajectory or a nonlinear asymptotic exponent. The initial phase retains spatial information and is generally different from Πu_0. After shrinking the initial basin so that its evolved points remain in the phase domain, uniqueness also gives $\Gamma(u(t)) = \phi_t \Gamma(u_0)$. Nonpolystable escape to a graded boundary is outside this metric basin around an HE representative.

12. The harmonic quotient metric and unitary rigidity

We now vary the holomorphic bundle, while keeping the smooth base fibre fixed. The resulting harmonic metric has a different domain from the metric in (113). For a skew-Hermitian connection one-form use $\|a\|_2^2 = -\int \mathrm{tr}(a \wedge *a)$, with volume $\omega_p^3/3!$. On the matrix space $W = \mathrm{Mat}_2 \oplus \mathrm{Mat}_2$ put

$$G_{0,p}(q,q) = 2c_+(p)\|q_A\|_{\mathrm{HS}}^2 + 2c_-(p)\|q_B\|_{\mathrm{HS}}^2. \tag{129}$$

Let $\mathfrak{k}_0 = \mathrm{i}\mathfrak{h}$ and let Q also denote the split complementary projection on skew-Hermitian trace-free sections. The fundamental vector for the compact action is

$$\rho_w \xi = (\xi_+ A - A\xi_-,\ \xi_- B - B\xi_+).$$

Exact equivariance of the actual connection map gives

$$d\mathcal{A}_p(w)(\rho_w \xi) = -D_{\mathcal{A}_p(w)}\xi. \tag{130}$$

The ineffective scalar may equivalently be included in the finite gauge algebra.

Theorem 12.1. *Uniformly in the base and every small w, for all $q \in W$ and $v \in QH^1(\mathfrak{su}E^{\mathrm{sp}})$,*

$$\begin{aligned}(1 - C|w|)\{G_{0,p}(q,q) + \|D_0 v\|_2^2\} &\le \|d\mathcal{A}_p(w)q - D_{\mathcal{A}_p(w)}v\|_2^2 \\ &\le (1 + C|w|)\{G_{0,p}(q,q) + \|D_0 v\|_2^2\}.\end{aligned} \tag{131}$$

Consequently the full unitary-gauge minimum is relatively equivalent, with error $O(|w|)$, to

$$g^{\mathrm{quot}}_{0,p,w}(q) = \inf_{\xi \in \mathfrak{k}_0} G_{0,p}(q - \rho_w \xi, q - \rho_w \xi). \tag{132}$$

Its kernel is exactly $\operatorname{im}\rho_w$. *For $F_p(z) = \mathcal{A}_p(\chi_p(z))$ and genuine tangents to a smooth orbit-type stratum of $\mu_{2,p}^{-1}(0)$, the actual harmonic connection metric satisfies*

$$(1 - C|z|)g^{\mathrm{quot}}_{0,p,z} \le F_p^* g_{\mathrm{harm}} \le (1 + C|z|)g^{\mathrm{quot}}_{0,p,z}. \tag{133}$$

This real metric is twice the Dolbeault harmonic Kodaira–Spencer metric. The estimates allow every change of stabilizer and all positive singular-value ratios of ρ_z.

Proof. Split harmonicity and type orthogonality give

$$\|a_{1,p}(q)\|_2^2 = 2\|\eta_p(q)\|_2^2 = G_{0,p}(q,q), \qquad \langle a_{1,p}(q), D_0 v\rangle_2 = 0.$$

The connection expansion (60) and the split section gap imply

$$\begin{aligned}\|d\mathcal{A}_p(w)q - D_{\mathcal{A}_p(w)}v - (a_{1,p}(q) - D_0 v)\|_2 &\le \|d\mathcal{N}_p(w)q\|_2 + \|[a_p(w), v]\|_2 \\ &\le C|w|\{G_{0,p}(q,q) + \|D_0 v\|_2^2\}^{1/2}.\end{aligned}$$

Squaring proves (131); density extends it to the stated H^1 domain. The minimum over QH^1 exists uniquely by coercivity. Infimizing the lower bound and taking $v = 0$ for the upper bound gives a relative comparison with $G_{0,p}(q,q)$.

Every full gauge direction is $\xi + v$. By (130), its residual is $d\mathcal{A}_p(w)(q + \rho_w\xi) - D_{\mathcal{A}_p(w)}v$. Apply the preceding comparison to every $q + \rho_w\xi$ and then infimize in ξ. Changing its sign gives (132). The finite range is closed at each parameter, so the kernel is precisely $\operatorname{im}\rho_w$. No estimate for the size of a minimizing ξ has been used.

Real equivariance gives $d\chi_p(z)\rho_z\xi = \rho_{\chi_p(z)}\xi$. Since $d\chi_p = I + O(|z|)$ in the uniformly positive arrow metric, the same comparison transfers through χ_p inside every residual before minimization. This proves (133) as a quotient form.

For its harmonic interpretation, take a smooth curve in one zero-level orbit-type stratum. Its actual derivative a at the HE connection D_A satisfies $(D_A a)^{0,2} = 0$ and $\mathrm{i}\Lambda D_A a = 0$. The full gauge minimum supplies the Coulomb representative $a_h = a - D_A\zeta$, with $D_A^* a_h = 0$. Subtracting this vertical preserves the two linearized equations because $F_A^{0,2} = 0$ and $\mathrm{i}\Lambda F_A = 0$. The Kähler identities identify the Coulomb and contracted equations with the sum and difference of the two adjoint expressions, giving

$$\bar\partial_A(a_h^{0,1}) = 0, \qquad \bar\partial_A^*(a_h^{0,1}) = 0.$$

This is the harmonic representative of the actual Kodaira–Spencer class; a unitary vertical is a Dolbeault coboundary. Type orthogonality gives the factor two in its real norm, in agreement with [Kob87, Section 7.2 and equations (7.6.5), (7.6.20)–(7.6.21)]. The orbit-type pieces of the zero level and their quotients by K are smooth by [SL91, Theorem 2.1]. The form is positive on their quotient tangents by its identified kernel. For fixed p it is smooth on each stratum, by the elliptic inverse on the complement of the constant-dimensional parallel kernel. No uniform derivative across strata or through the singular base is asserted. □

The same action occurs in the weak endomorphism energy:

$$E_{p,z}(S) = G_{0,p}(\rho_z(\mathrm{i}S), \rho_z(\mathrm{i}S)). \tag{134}$$

Thus its kernel is the infinitesimal stabilizer, while the null directions of the quotient tangent metric are the image of the action. These are different spaces related by one specified map.

12.1. **All unitary intertwiners are constant.** An infinitesimal metric comparison does not identify different finite quotient points with different bundle classes. The following statement supplies that identification.

Theorem 12.2. *On a common small parameter ball, every unitary gauge transformation intertwining $\mathcal{A}_p(w_1)$ and $\mathcal{A}_p(w_2)$ is a split parallel block unitary $K \in U(2) \times U(2)$, and $w_2 = K \cdot w_1$. At HE points, two constructed holomorphic bundles are isomorphic if and only if their parameters lie in the same K-orbit. The same assertion holds in the balanced z coordinates.*

Proof. Put $\rho = \max(|w_1|, |w_2|)$ and abbreviate the connection increments by $a_j = a_p(w_j)$. The intertwining equation gives

$$D_0 k = k a_1 - a_2 k, \qquad \|D_0 k\|_\infty \le C\rho.$$

Equation (97) holds for every small slice parameter: its proof did not use the HE equation. Taking divergence therefore gives

$$L_0 k = k D_0^* a_1 - (D_0^* a_2) k - \sum_j (D_{0,e_j} k) a_1(e_j) + \sum_j a_2(e_j) D_{0,e_j} k = O_\infty(\rho^2).$$

Let $\widehat{\Pi}$ project onto the full complex split parallel algebra $\mathrm{Mat}_2 \oplus \mathrm{Mat}_2$. The full complex inverse in Lemma 5.1 implies

$$\|k - \widehat{\Pi} k\|_{C^1(D_0)} \le C\rho^2. \tag{135}$$

Since k is unitary, $B = \widehat{\Pi} k$ satisfies $B^* B = I + O(\rho^2)$. Its block polar factor K_0 is unitary and $\|K_0^{-1} k - I\|_{C^1(D_0)} \le C\rho^2$. In the finite skew-Hermitian block algebra solve

$$\widehat{\Pi} \log(e^{-C} K_0^{-1} k) = 0.$$

The derivative in C at the identity is $-I$ and differs from it by $O(\rho^2)$ for these data. A uniform finite contraction gives $|C| \le C\rho^2$. Thus

$$k = K e^{\xi}, \qquad K = K_0 e^{C} \in K, \qquad \widehat{\Pi}\xi = 0, \qquad \|\xi\|_{C^1(D_0)} \le C\rho^2. \tag{136}$$

The determinant connection is fixed, so $\det k$ is constant. The small logarithm makes $\operatorname{tr}\xi$ constant, and its zero parallel projection makes this trace zero.

Conjugate the second connection by K^{-1} and put $\delta = K^{-1} \cdot w_2 - w_1$. The segment between these two parameters stays in the small slice, giving

$$a(K^{-1} \cdot w_2) - a(w_1) = a_{1,p}(\delta) + R_\delta, \qquad \|R_\delta\|_2 \le C\rho|\delta|.$$

The exact exponential gauge identity is

$$e^{\xi}D(w_1)e^{-\xi} - D(w_1) = -\int_0^1 e^{t\xi}(D(w_1)\xi)e^{-t\xi}\,\mathrm{d}t.$$

The split gap on $\widehat{\Pi}^{\perp}$, together with (136), therefore gives

$$\|a_{1,p}(\delta) + D_0\xi\|_2 \le C\rho(|\delta| + \|D_0\xi\|_2).$$

The two terms on the left are orthogonal, and their squared norm is $G_{0,p}(\delta,\delta) + \|D_0\xi\|_2^2$. The uniform positive arrow masses allow absorption for small ρ. Hence $\delta = 0$, $D_0\xi = 0$, and then $\xi = 0$. Every intertwiner is the constant K.

For a holomorphic isomorphism T between two HE members, the induced Hom connection has zero mean curvature. Its section Kähler identity gives $\|D_{\mathrm{Hom}}T\|_2^2 = 2\|\bar\partial_{\mathrm{Hom}}T\|_2^2 = 0$; see also [Kob87, Theorem 3.1.9(i)]. Thus T and its positive polar factor are parallel, and its unitary polar factor is an intertwiner. The first assertion applies, without simplicity or stability assumptions. The converse is equivariance. Finally injectivity and equivariance of χ_p transfer the assertion to balanced parameters. □

12.2. **The topology of the actual local locus.** On a fixed smooth fibre take the affine space of continuous unitary connections, with its C^0 topology, modulo C^1 unitary gauge. This quotient is Hausdorff. Indeed, suppose $D_{j,n} \to D_j$ and $D_{2,n} = k_n D_{1,n} k_n^{-1}$. The intertwiners are pointwise unitary and

$$D_0 k_n = k_n(D_{1,n} - D_0) - (D_{2,n} - D_0)k_n.$$

Their first derivatives are bounded on this fixed compact manifold. Arzelà–Ascoli gives a uniformly convergent subsequence; the equation then gives convergence of the first derivatives and a C^1 unitary intertwiner in the limit. Hence the orbit relation is closed. The orbit projection is open because saturations are unions of translates, so the quotient is Hausdorff. For smooth endpoints the displayed equation bootstraps an intertwiner to smoothness, introducing no extra identifications.

For any compact invariant subset of the balanced zero set, Theorem 12.2 makes its quotient map into actual connection classes continuous and injective. A compact-to-Hausdorff injection is a homeomorphism onto its image. These points are precisely distinct actual holomorphic bundle classes. The local completeness proved in Section 7 shows that a nested such locus contains representatives of every sufficiently small polystable object in the specified local germ. This is the germ coverage already proved there; χ_p is a real parametrization and does not supply a holomorphic chart or an algebraic Cartesian square of moduli spaces.

13. Local distance, completion and the metric tangent cone

Normalize the arrows by $T_p(A,B) = (\sqrt{c_+}A, \sqrt{c_-}B)$. The balanced set is

$$\mathcal{Z} = \{(X,Y) : XX^* = Y^*Y,\ YY^* = X^*X\},$$

with real metric $2(\|\mathrm{d}X\|_{\mathrm{HS}}^2 + \|\mathrm{d}Y\|_{\mathrm{HS}}^2)$. Let $\mathsf{W} = (\mathbb{Z}/2)^2 \rtimes S_2$ act on $\mathbb{C}^2$ by independent signs and permutation. For $u = (u_1,u_2)$ and $U = \mathrm{diag}(u_1,u_2)$, define

$$\mathfrak{F}_p([u]) = [\mathcal{A}_p(\chi_p(T_p^{-1}(U,U)))]. \tag{137}$$

The brackets on the right denote actual unitary gauge classes, equivalently actual holomorphic bundle classes at these HE points. Choose one $R_0 > 0$ so that $|u| \le R_0$ lies strictly inside every construction domain, uniformly in p. Put $\mathcal{M}_p(R) = \mathfrak{F}_p(\overline{B}_R/\mathsf{W})$ for $0 < R \le R_0$.

An admissible path in this locus is continuous and has a finite subdivision into smooth paths in its quotient orbit-type strata, with improper endpoint lengths allowed. Its length is the integral of the actual harmonic connection norm of its bundle-deformation tangent. Let $d_{p,R}$ be the infimum of these lengths among paths confined to $\mathcal{M}_p(R)$.

Theorem 13.1. *The map (137) is a homeomorphism of $\overline{B}_R/\mathsf{W}$ onto its actual image, with smooth image-stratum parametrizations. On every stratum its harmonic metric satisfies*

$$(1 - C|u|)\, 4|\mathrm{d}u|^2 \le \mathfrak{F}_p^* g_{\mathrm{harm}} \le (1 + C|u|)\, 4|\mathrm{d}u|^2. \tag{138}$$

For $|u|, |v| \le R$, set $d_{\mathrm{flat}}([u],[v]) = 2\min_{\gamma\in\mathsf{W}} |u - \gamma v|$. Then

$$\sqrt{1 - CR}\, d_{\mathrm{flat}}([u],[v]) \le d_{p,R}(\mathfrak{F}_p[u], \mathfrak{F}_p[v]) \le \sqrt{1 + CR}\, d_{\mathrm{flat}}([u],[v]). \tag{139}$$

The open-radius locus completes isometrically to this closed-radius locus. On the unit quotient ball,

$$\sup_{[u],[v]} \left|\rho^{-1} d_{p,\rho}(\mathfrak{F}_p[\rho u], \mathfrak{F}_p[\rho v]) - d_{\mathrm{flat}}([u],[v])\right| \le C\rho. \tag{140}$$

The same estimate holds with the single fixed-locus distance d_{p,R_0} for $\rho < R_0/8$. The actual local metric space therefore has the unique pointed metric tangent cone

$$(\mathbb{C}^2, 4|\mathrm{d}u|^2)/\mathsf{W} \tag{141}$$

at the split bundle. These comparisons and the resulting Gromov–Hausdorff convergence are uniform over the smooth conifold base, including simultaneous approaches to its discriminant. The distances use paths in the specified local locus.

13.1. **The exact finite quotient.** The balanced normal form in Section 7 is $X = \mathrm{diag}(d_j)$, $Y = \mathrm{diag}(d_j e^{\mathrm{i}\theta_j})$. Choose $u_j = d_j e^{\mathrm{i}\theta_j/2}$. Acting by $k_+ = \mathrm{diag}(e^{\mathrm{i}\theta_j/2})$, $k_- = I$ makes both arrows equal U. The phase at a zero entry is immaterial. Under the compact action XY changes by conjugation. For (U,U) its eigenvalue multiset is $\{u_1^2, u_2^2\}$. Thus two section points can be equivalent only if they differ by W. Conversely permutations are realized simultaneously in both blocks, and each sign change by a diagonal sign in the plus block alone. This proves the quotient identification, including repeated and zero entries. On a closed radius ball it is a homeomorphism by compactness, since $\|X\|_{\mathrm{HS}}^2 + \|Y\|_{\mathrm{HS}}^2 = 2|u|^2$ and the compact group quotient is Hausdorff.

The four strata are distinct nonzero squares, repeated nonzero squares, one zero square, and the origin. Their preimages are the complement of the four complex reflecting hyperplanes, the nonzero parts of $u_1 = \pm u_2$, the nonzero parts of the coordinate axes, and the origin. They agree with the actual factor and stabilizer types.

The section also gives smooth local coordinates on each stratum. On the balanced set XY is normal and $XX^* = |XY|$, since $(XY)(XY)^* = (XX^*)^2 = (XY)^*(XY)$. For distinct eigenvalues its spectral projections vary smoothly. For a nonzero eigenvalue λ, choose a smooth local square root u and a smooth unit eigenvector e_+; then $e_- = X^* e_+/\bar{u}$ is unit and satisfies $Xe_- = ue_+$, $Ye_+ = ue_-$. At a simple zero eigenvalue use smooth frames of the kernels, on which both arrows vanish. On the repeated nonzero stratum $XY = \lambda I$ and the same formula works with an arbitrary smooth plus frame. These are qualitative coordinates within a stratum; no uniform bound for their derivatives near another stratum is used.

For a diagonal section tangent (V,V) the ambient norm is $4\|V\|_{\mathrm{HS}}^2$. Its pairing with a vertical is

$$2\,\mathrm{Re}\,\mathrm{tr}\, V^*(\xi_+ U - U\xi_- + \xi_- U - U\xi_+) = 2\,\mathrm{Re}\,\mathrm{tr}\, V^*[\xi_+ + \xi_-, U] = 0.$$

Thus these tangents are horizontal, and the stratum coordinates just proved exhaust the quotient tangents. The exact finite quotient metric is $4\sum_j |\mathrm{d}u_j|^2$. On the repeated stratum it is $8|\mathrm{d}u|^2$, and on the one-nonzero stratum it is $4|\mathrm{d}u|^2$; the origin has zero stratum tangent.

Its path distance on a closed ball is d_{flat}. Local lifts of stratum paths and the triangle inequality give the lower bound. For the upper bound choose a minimizing γ and project the straight segment from u to γv. Convexity keeps it in the ball. Each reflecting hyperplane meets the segment at at most one parameter unless it contains the segment; a finite subdivision therefore makes the projected segment admissible. Its length attains the stated lower bound.

13.2. **Actual harmonic lengths and completion.** Theorem 12.2 and the compact quotient argument of Section 12 identify the finite quotient homeomorphically with actual classes. Each image stratum is represented by a smooth family of integrable HE connections. Its harmonic tangent has no kernel beyond the compact vertical directions by Theorem 12.1. Since $|T_p^{-1}(U,U)| \le C|u|$, that theorem and the exact horizontal metric prove (138). Integrating it over any admissible path gives the lower bound in (139); using the projected straight segment gives the upper bound. In particular this is a finite nondegenerate distance with exactly the compact quotient topology.

For interior endpoints the open-locus path distance equals the restriction of the closed-locus distance. One inequality is immediate. For the other, contract any finite-length closed-locus path by $u \mapsto (1-\epsilon)u$. The contracted pieces remain in their strata. The actual tensor is smooth there, so their speeds converge pointwise. Equation (138) dominates them by a constant times the original Euclidean speed, which is integrable by the lower bound in that equation. Dominated convergence proves convergence of lengths, including improper endpoint integrals. Add short radial paths at the two interior endpoints; their lengths tend to zero. Infimizing gives equality of the distances. Radial contraction also proves density of the open locus in the complete compact closed locus, establishing the claimed completion.

Homogeneity of d_{flat} and (139) with $R = \rho$ prove (140). The flat unit quotient ball has diameter at most four. The identity correspondence therefore has distortion $O(\rho)$, giving the uniform Gromov–Hausdorff bound directly.

13.3. **Localization in a fixed local metric space.** Shrink R_0 so that $CR_0 < 1/2$. For endpoints of parameter norm at most ρ, the straight-path upper bound is less than 5ρ. The radius $|u|$ is well defined on the quotient, and its variation is at most half the finite length. Hence a path starting at radius at most ρ and reaching 8ρ has actual length at least

$$2\sqrt{1-CR_0}(8\rho-\rho) > 9\rho.$$

Every path of length at most 6ρ between these endpoints stays inside radius 8ρ. Near-minimizing paths can be chosen with this length bound. Along them the lower tensor factor is $1-8C\rho$, while the straight segment supplies the upper factor $1+C\rho$. Thus the fixed-locus distance satisfies the same relative comparison with error $O(\rho)$ for these endpoints. This proves its version of (140).

To pass from parameter balls to actual metric balls, first the coarse comparison puts every small metric ball inside a comparable parameter ball. The sharper fixed-locus estimate then gives

$$d_{p,R_0}(0,\mathfrak{F}_p[u]) = 2|u|(1+O(|u|)).$$

An actual ball of radius 2ρ lies between parameter balls of radii $\rho(1-C'\rho)$ and $\rho(1+C'\rho)$. Radial projection between these changes distances by $O(\rho^2)$. After rescaling, the distortion is

$O(\rho)$. The same argument on every fixed multiple of ρ gives compatible pointed convergence to the full cone (141), proving uniqueness of the tangent.

Equivalently this cone is the symmetric square of $(\mathbb{C}, 4|\mathrm{d}u|^2)/\{\pm 1\}$, whose cone angle is π. The invariant coordinates $\lambda_j = u_j^2$ do not carry an ordinary Euclidean metric at zero. The theorem includes all local polystable factor types and every simultaneous admissible base approach. It concerns the harmonic metric on this actual local locus; it does not compare shortcuts through a larger moduli space or assert a metric on a singular base fibre.

References

[AHR20] J. Alper, J. Hall and D. Rydh, *A Luna étale slice theorem for algebraic stacks*, Ann. of Math. (2) **191** (2020), 675–738. doi:10.4007/annals.2020.191.3.1. Locators refer to arXiv:1504.06467v3.

[ACG13] L. Álvarez-Cónsul, M. Garcia-Fernandez and O. García-Prada, *Coupled equations for Kähler metrics and Yang–Mills connections*, Geom. Topol. **17** (2013), 2731–2812. doi:10.2140/gt.2013.17.2731. Locators refer to arXiv:1102.0991v3.

[BS21] N. Buchdahl and G. Schumacher, *An analytic application of Geometric Invariant Theory*, J. Geom. Phys. **165** (2021), 104237. doi:10.1016/j.geomphys.2021.104237. Locators refer to arXiv:2008.04625v3.

[Car52] H. Cartan, *Faisceaux analytiques sur les variétés de Stein*, Séminaire Henri Cartan **4** (1951–1952), exposé 18; *Faisceaux analytiques sur les variétés de Stein: démonstration des théorèmes fondamentaux*, exposé 19. Exposé 18, exposé 19.

[CLZZ26] Z. Chen, C. Li, C. Zhang and X. Zhang, *Long-time behavior of the Hermitian–Yang–Mills flow on non-Kähler manifolds*, preprint, January 9, 2026. arXiv:2601.05614v1.

[Chu12] M.-T. Chuan, *Existence of Hermitian–Yang–Mills metrics under conifold transitions*, Comm. Anal. Geom. **20** (2012), 677–749. doi:10.4310/CAG.2012.v20.n4.a1. Locators refer to arXiv:1012.3107v1.

[CPY24] T. C. Collins, S. Picard and S.-T. Yau, *Stability of the tangent bundle through conifold transitions*, Comm. Pure Appl. Math. **77** (2024), 284–371. doi:10.1002/cpa.22135. Locators refer to arXiv:2102.11170v2.

[dCFI26] M. A. de Cataldo, A. Fernandez Herrero and A. Ibáñez Núñez, *Relative étale slices and cohomology of moduli spaces*, Ann. Inst. Fourier, online first (2026). doi:10.5802/aif.3794. Locators refer to arXiv:2307.00350v3.

[Del25] R. Delloque, *Perturbations of a Vector Bundle whose Curvature Form Solves a Polynomial Equation*, preprint, July 2, 2025. arXiv:2507.01534v1.

[Dem12] J.-P. Demailly, *Complex Analytic and Differential Geometry*, version of June 21, 2012. Author's text.

[FK01] T. Friedrich and E. C. Kim, *Some remarks on the Hijazi inequality and generalizations of the Killing equation for spinors*, J. Geom. Phys. **37** (2001), 1–14. Locators refer to arXiv:math/9906168v1.

[GL83] M. Gromov and H. B. Lawson, Jr., *Positive scalar curvature and the Dirac operator on complete Riemannian manifolds*, Publ. Math. IHÉS **58** (1983), 83–196. doi:10.1007/BF02953774.

[HKKP18] F. Haiden, L. Katzarkov, M. Kontsevich and P. Pandit, *Iterated logarithms and gradient flows*, preprint, February 12, 2018. arXiv:1802.04123v1.

[HKKP23] F. Haiden, L. Katzarkov, M. Kontsevich and P. Pandit, *Semistability, modular lattices, and iterated logarithms*, J. Differential Geom. **123** (2023), 21–66. doi:10.4310/jdg/1679503804. Locators refer to arXiv:1706.01073v2.

[HL97] D. Huybrechts and M. Lehn, *The Geometry of Moduli Spaces of Sheaves*, first edition, 1997. Locators refer to the first-edition text.

[JSZ08] D. Jakobson, A. Strohmaier and S. Zelditch, *On the spectrum of geometric operators on Kähler manifolds*, J. Mod. Dyn. **2** (2008), 701–718. doi:10.3934/jmd.2008.2.701. Locators refer to the author's preprint of May 19, 2008.

[Kin94] A. D. King, *Moduli of representations of finite dimensional algebras*, Quart. J. Math. Oxford (2) **45** (1994), 515–530. doi:10.1093/qmath/45.4.515.

[Kle05] S. L. Kleiman, *The Picard scheme*, in *Fundamental Algebraic Geometry: Grothendieck's FGA Explained*, Mathematical Surveys and Monographs, vol. 123, American Mathematical Society, Providence, RI, 2005, 235–321. Locators refer to arXiv:math/0504020v1.

[Kob87] S. Kobayashi, *Differential Geometry of Complex Vector Bundles*, Publications of the Mathematical Society of Japan, vol. 15, 1987. Locators refer to the publisher's 2013 re-typesetting.

[Lam26] S. T. Lam, *Semistability and asymptotics of geometric flows*, preprint, version of June 8, 2026. arXiv:2606.09235v1.

[LR18] R. Lärkäng and J. Ruppenthal, *Koppelman formulas on affine cones over smooth projective complete intersections*, Indiana Univ. Math. J. **67** (2018), 753–780. doi:10.1512/iumj.2018.67.6309. Locators refer to arXiv:1509.00987v2.

[Led11] M. Ledoux, *From concentration to isoperimetry: semigroup proofs*, Contemp. Math. **545** (2011), 155–166. Author's offprint.

[LM06] A. Legrand and S. Moroianu, *On the L^p index of spin Dirac operators on conical manifolds*, Studia Math. **177** (2006), 97–112. doi:10.4064/sm177-2-1. Locators refer to arXiv:math/0407027v1.

[LY86] P. Li and S.-T. Yau, *On the parabolic kernel of the Schrödinger operator*, Acta Math. **156** (1986), 153–201. doi:10.1007/BF02399203.

[Sim94] G. Simonett, *Invariant manifolds and bifurcation for quasilinear reaction-diffusion systems*, Nonlinear Anal. **23** (1994), 515–544. Author-hosted scan.

[SL91] R. Sjamaar and E. Lerman, *Stratified symplectic spaces and reduction*, Ann. of Math. (2) **134** (1991), 375–422. Locators refer to the author manuscript.

[Sta] The Stacks Project Authors, *The Stacks Project*, stacks.math.columbia.edu. Individual tags are given at each use.

[Tod18] Y. Toda, *Moduli stacks of semistable sheaves and representations of Ext-quivers*, Geom. Topol. **22** (2018), 3083–3144. arXiv:1710.01841v3.

[W26a] B. J. Wuebben, *Nonsimplicial local stability chambers of toric tangent bundles*, preprint, October 29, 2026. doi:10.5281/zenodo.23774770.

[W26b] B. J. Wuebben, *Integral topology and small eigenvalues in Calabi–Yau conifold transitions*, preprint, November 9, 2026. doi:10.5281/zenodo.24096887.

[W26c] B. J. Wuebben, *Relative Hodge spectra of multiscale conifold smoothings*, manuscript, November 12, 2026. doi:10.5281/zenodo.24175848.

[W26d] B. J. Wuebben, *Quantitative metric wall crossing for K3 bundle moduli*, preprint, November 6, 2026. doi:10.5281/zenodo.24024998.

[W26e] B. J. Wuebben, *Quantitative local models for Hermitian–Einstein metrics*, preprint, November 7, 2026. doi:10.5281/zenodo.24029090.

[W26f] B. J. Wuebben, *Uniform asymptotic phase for Hermitian–Yang–Mills flow near a polystable bundle*, preprint, November 7, 2026. doi:10.5281/zenodo.24038504.

[W26g] B. J. Wuebben, *All-time approximation of Hermitian–Yang–Mills flow on degenerating curves*, preprint, November 11, 2026. doi:10.5281/zenodo.24149212.